

7

NUMBER AND ALGEBRA

GRAPHING CURVES

When an object is thrown upwards, its path is a curve called a parabola. The shape and length of the path will depend on the initial speed of the object. Furthermore, car headlights and satellite dishes all use mirrors or reflectors that have the shape of a parabola.



Chapter outline

		Working mathematically				
7.01	Direct variation*	U	F	PS	R	C
7.02	Inverse variation*	U	F	PS	R	C
7.03	Conversion graphs*	U	F		R	C
7.04	Graphing quadratic equations $y = ax^2 + c$	U	F		R	C
7.05	Applying quadratic equations	U	F		R	C
7.06	Graphing hyperbolas $y = \frac{k}{x} + c$*	U	F		R	C
7.07	Graphing exponential equations $y = a^x + c$	U	F		R	C
7.08	Exponential growth and decay	U	F		R	C
7.09	Identifying graphs	U	F		R	C

*PATH

Note: See 14.05 for Graphing quadratic equations $y = ax^2 + bx + c$

Wordbank

asymptote A line that a curve gets very close to but never touches, for example, the x-axis is an asymptote of the exponential curve

direct variation or **direct proportion** A relationship between 2 variables of the form $y = kx$, where k is a constant; for example, if $y = 8.5x$, then y is directly proportional to x

exponential equation An equation involving a variable as a power, such as $y = 3^x$, whose graph is an exponential curve

inverse variation or **inverse proportion** A relationship between 2 variables of the form $y = \frac{k}{x}$, where k is a constant; for example, if $y = \frac{50}{x}$, then y is inversely proportional to x

parabola A U-shaped curve that is the graph of a quadratic equation

quadratic equation An equation involving a variable squared (power of 2), such as $y = 3x^2 - 6$, whose graph is a curve called a parabola

Quiz
Wordbank 7

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (8):
- **7.01** Direct proportion • Direct linear variation
- **7.02** Inverse variation • Inverse proportion
- **7.04** The parabola $y = ax^2 + c$
- **7.05** Applying quadratic functions
- **7.06** Graphing hyperbolas
- **7.07** The exponential curve

***Twig* videos (2):**
- **7.02** The heartbeat formula
- **7.07** The emperor's chessboard

***PhET* interactive (1):**
- **7.04** Graphing quadratics

Quizzes (5):
- Wordbank 7
- SkillCheck 7
- Mental skills 7
- Language of maths 7
- Test yourself 7

Worksheets (7):
- **7.02** Direct and inverse proportion • Variation problems
- **7.03** Currency conversion graph
- **7.04** Graphing parabolas
- **7.06** Graphing hyperbolas
- **7.07** Graphing exponentials
- **Mind map** Graphing curves

Puzzles (2):
- **7.09** Matching graphs
- **Language of maths** Graphing curves crossword

Technology (2):
- **7.04** Investigating parabolas 1
- **7.04, 7.06** Curve sketcher

Spreadsheets (2):
- **7.04** Investigating parabolas 1
- **7.04, 7.06** Curve sketcher

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- (PATH) solve problems involving direct and inverse variation
- (PATH) read and interpret conversion graphs
- graph parabolas of the form $y = ax^2$ and $y = ax^2 + c$
- graph exponential curves of the form $y = a^x$
- solve problems that can be modelled by quadratic and exponential equations
- (PATH) graph hyperbolas of the form $y = \frac{k}{x}$
- match graphs to their equations
- use technology to find points of intersection between a line and a curve

Quiz
SkillCheck 7

SkillCheck ANSWERS ON P. 598

1 If $A = 2x^2 - 3$, find A if:

a $x = 1$ **b** $x = 4$ **c** $x = 0$ **d** $x = -6$

2 If $y = 5^x$, find y if:

a $x = 4$ **b** $x = 5$ **c** $x = 0$ **d** $x = -2$

3 If $S = \frac{120}{t}$, find S if:

a $t = 10$ **b** $t = 35$ **c** $t = 2.5$ **d** $t = 1.8$

Direct variation 7.01

Two variables are **directly proportional** to each other if one variable is a constant multiple of the other; when one variable changes, the other one changes by the same factor.
This relationship is called **direct variation** or **direct proportion**.

PATH

7.01

Videos
Direct proportion
Direct linear variation

Direct variation

If y is **directly proportional** to x, written as '$y \propto x$', then $y = kx$, where k is a constant (number) called the **constant of variation** or **constant of proportionality**.

- A direct linear relationship exists between x and y
- If x increases (or decreases), y increases (or decreases)
- If x is doubled (or halved), y is doubled (or halved)
- Another way of saying 'y is directly proportional to x' is 'y **varies directly** as x'
- The graph of direct variation is a straight line going through (0, 0) with gradient k

Example 1

The distance (d) in metres travelled by a cyclist on a road bike is directly proportional to the number of rotations (r) of its wheels. After 750 rotations, a distance of 1600 m is travelled.

a What distance will be travelled after 1200 rotations?

b How many full rotations will be needed to cover 30 km?


iStock.com/torwai

Solution

a d is directly proportional to r, or $d \propto r$.

$\therefore d = kr$ Forming a variation equation

To find k, substitute the information given for r and d.

When $r = 750$, $d = 1600$:

$1600 = k \times 750$ Finding k

$k = \frac{1600}{750}$

$k = 2.133...$

$\therefore d = 2.133... r$ Rewriting the equation $d = kr$

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When $r = 1200$, Solving the problem

$d = 2.133... \times 1200$

$= 2560$ m

After 1200 rotations, the distance travelled will be 2560 m.

b When $d = 30$ km $= 30\,000$ m Distance must be in metres

$30\,000 = 2.133... \times r$

$r = \frac{30\,000}{2.133...}$

$= 14\,062.5$

$\approx 14\,063$ rotations Rounding up for full rotations

For a distance of 30 km, there will be 14 063 rotations.

Solving direct variation problems

1 Identify the 2 variables (say x and y) and form a variation equation, $y = kx$
2 Substitute values for x and y to find k, the constant of variation
3 Rewrite $y = kx$ using the value of k
4 Substitute a value for x or y into $y = kx$ to solve the problem

Example 2

M varies directly as n. If when $n = 6$, $M = 103.8$, find M when $n = 14.2$.

Solution

$M = kn$ Forming a proportion equation

To find k, substitute $n = 6$ and $M = 103.8$.

$103.8 = k \times 6$ Finding k

$6k = 103.8$

$k = \frac{103.8}{6}$

$= 17.3$

$\therefore M = 17.3n$ Rewriting the equation $M = kn$

Substitute $n = 14.2$ to find M. Solving the problem

$M = 17.3 \times 14.2$

$= 245.66$

EXERCISE 7.01 ANSWERS ON P. 598

PATH

Direct variation U F PS R C

EXAMPLE 1

7.01

1 Match each statement with its variation equation (k is the constant of variation). C

a The distance, D, travelled is directly proportional to the time, T.

b The wage, W, earned is directly proportional to the hours, h, worked.

c The wedding cost, C, varies directly as the number, n, of guests.

d The interest, I, earned varies directly as the size of the deposit, D

A $C = kn$ **B** $I = kD$ **C** $D = kT$ **D** $W = kh$

2 The distance, D, travelled by Connor, a marathon runner, varies directly as time, T. PS R C

Time, T (min)	1	2	3
Distance, D (m)	190	380	570

a Find the constant of variation and write the equation for D.

b How far, in kilometres, will Connor run in:

i 20 minutes? **ii** 45 minutes?

c How long would it take Connor to run 12.35 kilometres? Answer in hours and minutes.

3 Mehta's earnings for working a shift at the local nursery are directly proportional to the number of hours she works. Yesterday, she earned $304.30 for working an 8.5-hour shift. PS R C

a If Mehta's earnings are represented by E, and the number of hours worked is represented by h, find the constant of variation and write an equation for E.

b How much will she earn for working a 7-hour shift?

c How many hours did she work today if she earned $196.90 for the shift?



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4 The amount of interest, I, earned for one year on an investment account varies directly as the size of the deposit, D. PS R C

a If Caterina earns \$826.50 interest on an investment of \$14 250, find the variation equation for I, including the constant of variation.

b Hence, how much will she earn on an investment of \$9000?

c If she doubles the size of her investment in **b**, how much will she earn in interest?

EXAMPLE 2

5 S varies directly as t. If when $t = 14$, $S = 106.4$, what is the value of S when $t = 0.3$? Select the correct answer **A**, **B**, **C** or **D**. R C

A 2.28 **B** 27.72 **C** 36.12 **D** 446.88

6 Find the linear formula for b in terms of a for this table of values.

a	4	8	12	16	20
b	10	20	30	40	50

7 The line graph shows that the cost of hamburgers purchased from the local takeaway store depends directly on the number of burgers purchased. PS R C

a Copy this table and use the graph to complete it.

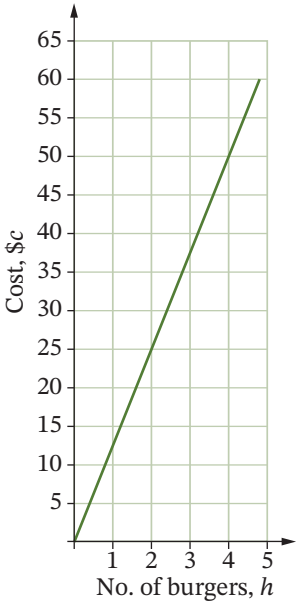
No. of burgers, h	1	2	3
Cost, c (\$)			

b Find the variation equation to represent the relationship between the cost (\$$c$) and the number of burgers (h).

c If Kendall buys 6 hamburgers, what is the total cost of the hamburgers?

d The total cost of one order of hamburgers is \$137.50. How many hamburgers were ordered?

e Find the gradient of the line. How is it related to the constant of variation?



8 K varies directly as L. If $L = 9.5$ when $K = 1045$, what is the value of K when $L = 1.65$? Select **A**, **B**, **C** or **D**. R C

A 0.015 **B** 93.7 **C** 181.5 **D** 1708.575

9 A linear relationship exists between the mass of a car (m kg) and its rate of fuel consumption (F L/100 km). PS R C

a Find the variation equation for F if a 1500 kg car uses fuel at a rate of 9 L/100 km.

b Find the fuel consumption of a 2500 kg car.

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10 For an object that is cooling, the drop in temperature varies directly as time. If the temperature drops 8°C in 5 minutes, how long would it take to drop 10°C? Select **A**, **B**, **C** or **D**. PS R C

A 6.25 min **B** 7 min **C** 12.8 min **D** 16 min

11 The weight of an astronaut on Mars is proportional to his weight on Earth. A 72 kg astronaut weighs 27.4 kg on Mars. PS R C

a Calculate how much a 60 kg astronaut weighs on Mars, correct to 1 decimal place.

b If an astronaut weighs 32 kg on Mars, calculate his weight on Earth, correct to 1 decimal place.



Esteban De Armas/Shutterstock.com

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7.02

Inverse variation 7.02

Two variables are **inversely proportional** to each other if when one variable increases, the other one decreases by the same factor. This relationship is called **inverse variation** or **inverse proportion**.

The table below shows the different speeds of a car (s km/h), and the time it takes to travel 100 km (t min). As the speed increases, the time taken decreases.

Speed (s km/h)	50	60	80	100
Time (t min)	120	100	75	60

Inverse variation

If y is **inversely proportional** to x, written as '$y \propto \frac{1}{x}$', then $y = \frac{k}{x}$, where k is a constant (number) called the **constant of variation** or **constant of proportionality**.

- If x increases, y decreases ('inverse' means 'opposite')
- If x decreases, y increases
- If x is doubled, y is halved
- If x is halved, y is doubled
- Another way of saying 'y is inversely proportional to x' is 'y **varies inversely** as x'

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Worksheets
Direct and inverse proportion

Variation problems

Videos
Inverse variation

The heartbeat formula

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Example 3

The time (t) in minutes taken by a car to travel 100 km is inversely proportional to the speed (s km/h) of the car, as shown in the table above. At 50 km/h, the time taken is 120 minutes.

a Find the constant of variation and the inverse variation equation for t.

b How long did the car take to travel 100 km at:

i 40 km/h? **ii** 110 km/h?

c Find the car's speed if it took 45 minutes to travel 100 km.

Solution

a t is inversely proportional to s.

$\therefore t = \frac{k}{s}$ Forming an inverse variation equation

To find k, substitute the information given for s and t.

When $s = 50$, $t = 120$: Finding k, the constant of variation

$120 = \frac{k}{50}$

$k = 120 \times 50$

$= 6000$

The constant of variation is 6000.

$\therefore t = \frac{6000}{s}$ Rewriting the equation $t = \frac{k}{s}$

b **i** When $s = 40$, $t = \frac{6000}{40} = 150$ min

At 40 km/h, the trip takes 150 min (or 2 h 30 min).

ii When $s = 110$,

$t = \frac{6000}{110}$

$= 54.5454...$

≈ 55 min

At 110 km/h, the trip takes 55 min.

c When $t = 45$,

$45 = \frac{6000}{s}$

$45s = 6000$

$s = \frac{6000}{45}$

$= 133\frac{1}{3}$ km/h

For a travel time of 45 min, the speed must be $133\frac{1}{3}$ km /h.

Solving inverse variation problems

1 Identify the 2 variables (say x and y) and form a variation equation, $y = \frac{k}{x}$

2 Substitute values for x and y to find k, the constant of variation

3 Rewrite $y = \frac{k}{x}$ using the value of k

4 Substitute a value for x or y into $y = \frac{k}{x}$ to solve the problem

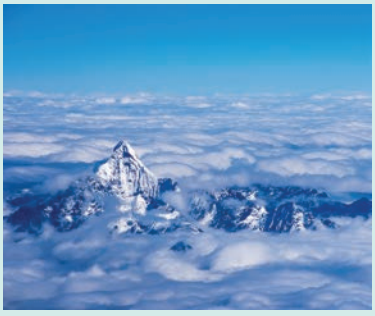
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Example 4

The temperature, T (in degrees Celsius), of the air is inversely proportional to the height, h (in metres), above sea level. At 600 m above sea level, the temperature is 8°C.

a What is the temperature at 1000 m above sea level?

b Graph the relationship between temperature and height above sea level, for heights between 0 and 5000.



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Video
Inverse proportion

7.02

Solution

a T is inversely proportional to h.

$T = \frac{k}{h}$ Forming a variation equation

Substitute $h = 600$ and $T = 8$ to find k.

$8 = \frac{k}{600}$ Finding k

$k = 8 \times 600$

$= 4800$

$\therefore T = \frac{4800}{h}$ Rewriting the equation $T = \frac{k}{h}$

When $h = 1000$, Solving the problem

$T = \frac{4800}{1000} = 4.8°C$

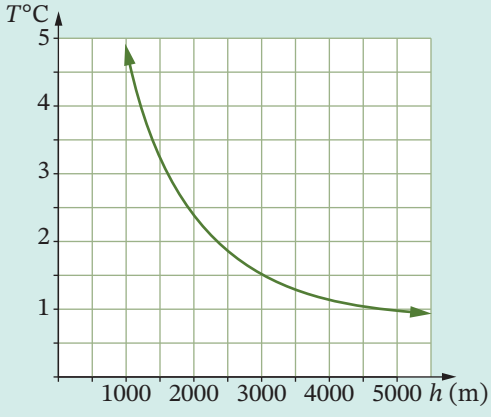
The temperature at a height of 1000 metres above sea level is 4.8°C.

b Draw a table of values for $T = \frac{4800}{h}$

h	1000	2000	3000	4000	5000
T	4.8	2.4	1.6	1.2	0.96

Note that as h increases, T decreases.

This graph is called a **hyperbola**.



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Example 5

P varies inversely as q. If when $q = 4$, $P = 38$, find P when $q = 5$.

Solution

$P = \frac{k}{q}$ — Forming a variation equation

Substitute $q = 4$, $P = 38$

$38 = \frac{k}{4}$ — Finding k

$k = 38 \times 4 = 152$

$\therefore P = \frac{152}{q}$ — Rewriting the equation $P = \frac{k}{q}$

Substitute $q = 5$ to find P. — Solving the problem

$P = \frac{152}{q} = 30.4$

EXERCISE 7.02 ANSWERS ON P. 598

Inverse variation U F PS R C

1 Match each statement with its inverse variation equation (k is the constant of variation). C

a The travel time, T, is inversely proportional to the speed, S.

b The temperature, T, above sea level varies inversely as the altitude, h.

c The cost, C, per person of hiring a bus varies inversely as the number, n, of people.

d The time, T, taken to move house is inversely proportional to the number, n, of helpers.

A $T = \frac{k}{n}$ **B** $T = \frac{k}{S}$ **C** $T = \frac{k}{h}$ **D** $C = \frac{k}{n}$

EXAMPLE 3

2 The time taken, T hours, to travel from Sydney to Melbourne varies inversely as its average speed, s km/h. PS R C

a If it takes 11.5 hours at an average speed of 80 km/h, find the constant of variation and the variation equation for T.

b If the average speed is increased to 90 km/h, how long will the journey take? Answer in hours and minutes.

c Find the average speed needed to complete the trip in 10 hours.

EXAMPLE 4

3 Air temperature, T (in degrees Celsius), varies inversely as its height, h (in metres), above sea level. At 150 m above sea level, the temperature is 30°C. PS R C

a Find the constant of variation and write a variation equation for T.

b What is the temperature at:

i 300 m above sea level? **ii** 2500 m above sea level?

c What is the height above sea level when the temperature is:

i 8°C? **ii** 22.5°C?

d Graph the relationship between temperature and height above sea level. Use T on the vertical axis and h on the horizontal axis with $h = 0, 500, 1000, 1500, ..., 3000$.

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PATH

7.02

4 The maximum number of diners, N, in Yen's restaurant is inversely proportional to the amount of floor space, S m^2, allocated to each diner. If 1.4 m^2 is allowed per diner, the restaurant can have 80 diners. PS R C

a Write a variation equation for N.

b How many diners could the restaurant contain if only 1.1 m^2 was allocated per person?

c How much space (correct to one decimal place) is allocated to each diner if the restaurant has a maximum capacity of 55 diners?

d During the COVID-19 pandemic of 2020, social distancing measures introduced to minimise the spread of the virus allocated 4 m^2 to each diner. How many diners could Yen's restaurant serve under this restriction?

5 The rate of vibration of a string varies inversely as its length. A string that is 330 mm long vibrates at 196 Hz (hertz). What length of string will vibrate at 500 Hz? Select the correct answer **A**, **B**, **C** or **D**. PS R C

A 129 mm **B** 297 mm **C** 323 mm **D** 842 mm

6 Which equation represents this table of values? Select **A**, **B**, **C** or **D**. R

x	2	5	8	10
y	2.5	1	0.625	0.5

A $y = \frac{10}{x}$ **B** $y = \frac{5}{x}$ **C** $y = \frac{2.5}{x}$ **D** $y = \frac{1}{x}$

7 K is inversely proportional to L. If $L = 2$ when $K = 7$, find K when $L = 15$. R

EXAMPLE 5

8 Ankit believes that at a train station, the number of people waiting on the platform is inversely proportional to the time until the next train arrives. According to his model, when there are 15 people waiting, the train will arrive in 4 minutes. PS R C

a When will the train arrive if there are 25 people waiting?

b How many people are waiting at the station 12 minutes before the train arrives?



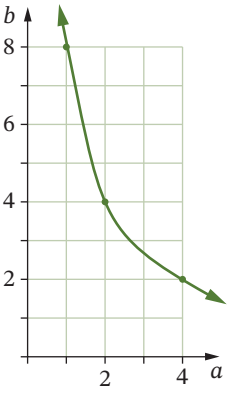
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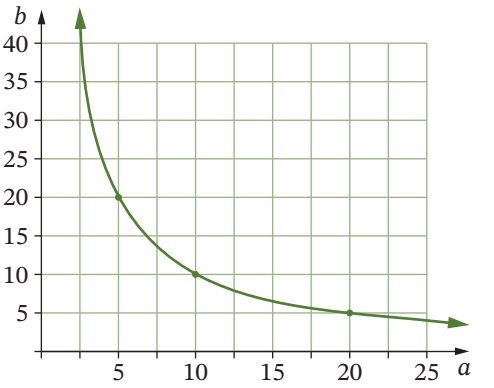
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9 Each graph shows an inverse relationship between a and b. Find each variation equation. R C

a



b



10 The time taken to travel from Sydney to Brisbane varies inversely as the average speed. A train travelling at an average speed of 70 km/h takes 14 h 7 min. How long would the same trip take if the average speed of the train was: PS R C

a 50 km/h? **b** 75 km/h?

11 The amount of time it takes Sonia to paint her house is inversely proportional to the number of friends she has to help her. When she has 5 friends helping, the job takes 4 days. PS R C

a How long will it take if she has 3 friends helping?

b How many friends must she have to help her to paint the house in 2 days?

12 The microwave oven cooking time (t seconds) for a meal varies inversely as the oven's power rating (W watts). For an 800 W microwave oven, the cooking time is 300 seconds. PS R C

a Find the variation equation for t.

b What is the cooking time for the meal in a 1200 W microwave oven?

c Find the power rating (to the nearest 100 W) of the microwave oven if the cooking time is:

i 6 min 20 s **ii** 2 min 40 s

13 The pressure (P atmospheres) of an amount of gas varies inversely as its volume (V litres). For a particular gas, its pressure is 5.2 atm if its volume is 4.5 L. PS R C

a Find the variation equation for P.

b Calculate the volume of the gas when its pressure is 2.8 atm.

c If its volume increases to 8 L, what is its pressure?

Conversion graphs

7.03

A **conversion graph** is used to convert from one unit to another; for example, miles to kilometres, or Australian dollars to US dollars. It usually contains one straight line that begins at the origin (0, 0), so it has an equation in the form $y = kx$ and is an example of **direct variation**.

PATH

Worksheet
Currency conversion graph

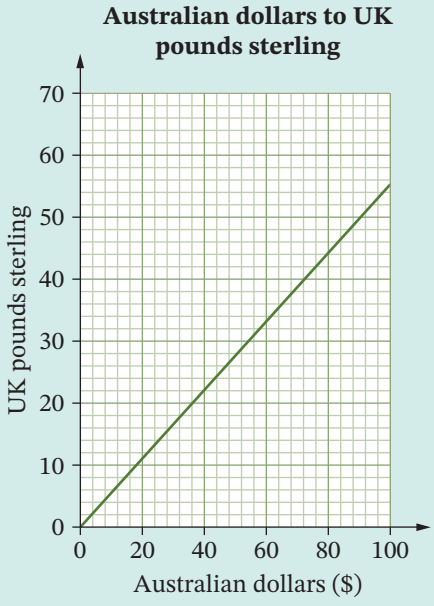
Example 6

Exchange rates change daily, but suppose the exchange rate between the Australian dollar and the UK pound sterling is \$A1 = £0.553, then \$A100 = £55.30 sterling. These values are used to draw this conversion graph.

Use the graph to convert:

a \$A50 to pounds

b £10 to Australian dollars.



Solution

Reading from the graph:

a \$A50 ≈ £28

b £10 ≈ \$A18

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EXERCISE 7.03 ANSWERS ON P. 599

Conversion graphs U F R C

EXAMPLE 6

1 Use the graph on the previous page for this question.

a Convert to pounds:

i $A40

ii $A88

b Convert to Australian dollars:

i £18

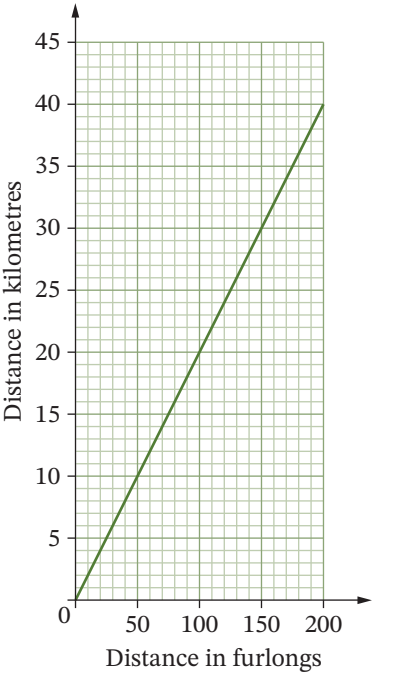
ii £60

c In 2015, $A1 = £0.50. R C

i How much less was $A40 worth in UK pounds sterling in 2015 than it is using this more recent conversion graph?

ii How much more was £60 worth in Australian dollars in 2015 for UK tourists visiting Australia, than it is using this more recent conversion graph?

2 The furlong is an imperial measure once used to measure length. This conversion graph shows distances in furlongs converted to kilometres.



a Convert to kilometres:

i 10 furlongs

ii 100 furlongs

iii 170 furlongs

b Convert to furlongs:

i 10 km

ii 25 km

iii 36 km

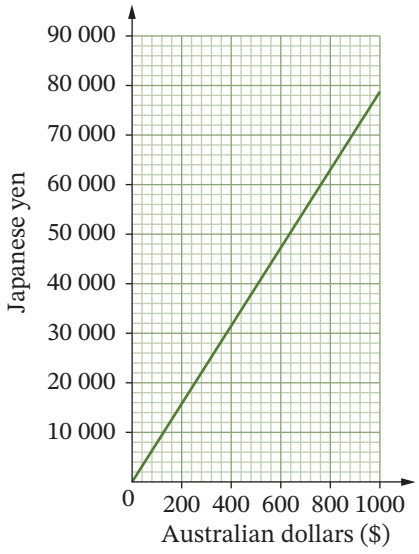
c Use an answer from part **a** to convert 300 furlongs to kilometres. R

d Use an answer from part **b** to convert 100 kilometres to furlongs. R

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3 This graph converts between Australian dollars and Japanese yen (¥).



a Convert to Japanese yen:

i $200

ii $800

iii $1000

b Convert to Australian dollars.

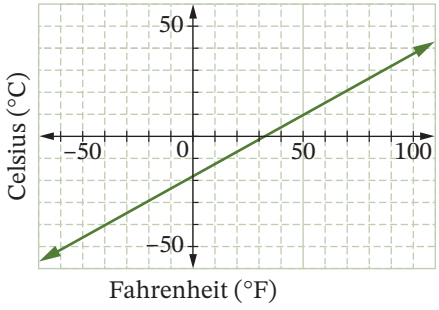
i ¥20 000

ii ¥60 000

iii ¥72 000

4 This graph converts temperatures between degrees Fahrenheit and degrees Celsius. Convert:

Degrees Fahrenheit to degrees Celsius



a 0°F to °C

b 50°F to °C

c 80°F to °C

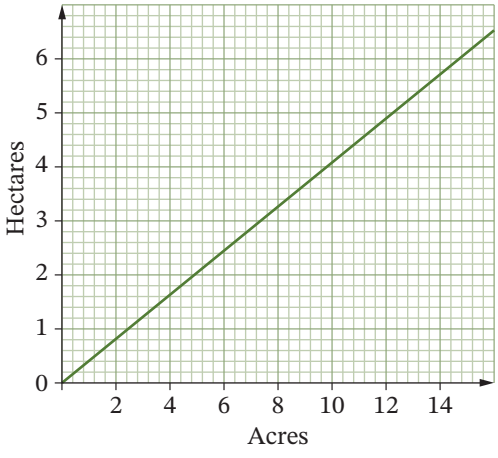
d 0°C to °F

e −10°C to °F

f 30°C to °F

5 This conversion graph converts between acres and hectares. The acre is an imperial measure of land area.

Converting acres to hectares



a Use the graph to convert 12 acres to hectares.

b A garden has an area of 5 acres. What is this area in hectares?

c Use the graph to convert 4.4 hectares to acres.

d Mr Ferguson has a property with an area of 5 hectares. How big is this in acres?

e A rectangular playing field measures 250 m by 128 m.

i What is the area of the field in square metres?

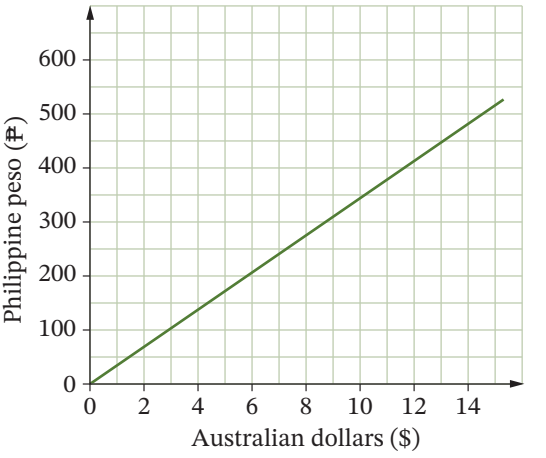
ii What is the area of the field in hectares?

iii What is the area of the field in acres?

Foundation | Standard | Complex

PATH

6 This graph is used to convert Australian dollars to Philippine pesos (₱).



a Change each Australian dollar amount into Philippine pesos.

i $15

ii $6

iii $9

b Change ₱350 to Australian dollars.

c How many Australian dollars would you receive for ₱200?

d Estimate the number of pesos you should get for $120. R

7.04 Graphing quadratic equations $y = ax^2 + c$

Worksheet Graphing parabolas

Technology Investigating parabolas 1

Curve sketcher

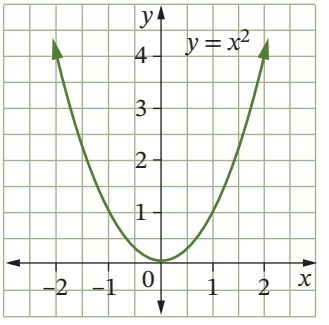
Spreadsheets Investigating parabolas 1

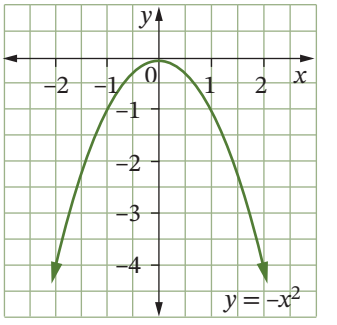
Curve sketcher

Interactive Graphing quadratics

An equation in which the highest power of the variable is 2 is called a **quadratic equation**, for example, $y = 2x^2 - 5$, $y = x^2 + 7x + 12$ and $y = -5x^2$. The graph of a quadratic equation is a smooth U-shaped curve called a **parabola**.

The graphs of $y = x^2$ and $y = -x^2$ are shown below.





Concave up (looks like a smile ☺)

Minimum value of the parabola is $y = 0$.

Concave down (looks like a frown ☹)

Maximum value of the parabola is $y = 0$.

Both parabolas have one axis of symmetry, the y-axis, and a turning point, called the **vertex**, at the origin (0, 0).

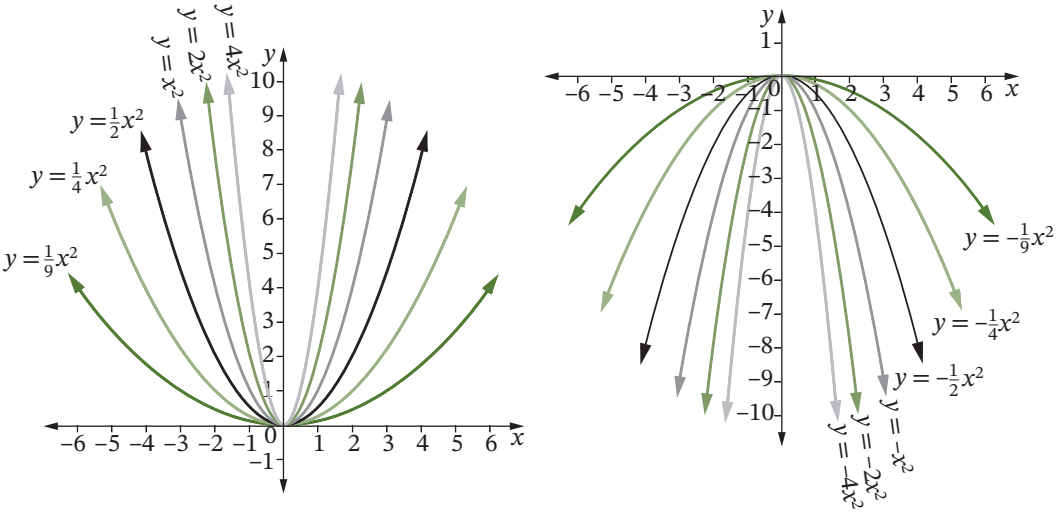
Foundation Standard Complex

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The graph of $y = ax^2$

For the graph of $y = ax^2$, where a is a constant (number), the size of a (the **coefficient** of x^2) affects whether the parabola is 'wide' or 'narrow'.

As the size of a increases, the parabola becomes 'narrower' and as the size of a decreases, the parabola 'widens'. If a is negative, then the parabola is concave down.



The graph of $y = ax^2$ is a parabola with the y-axis as its **axis** (of symmetry) and $(0, 0)$ as its **vertex**.



Neale Cousland/Shutterstock.com

The graph of $y = ax^2 + c$

For the graph of $y = ax^2 + c$, where a and c are constants, the effect of c is to move the parabola up or down from the origin. Also, c is the y-intercept of the parabola.

Video
The parabola
$y = ax^2 + c$

Example 7

Graph each set of quadratic equations on the same number plane, showing the vertex of each parabola.

a $y = x^2, y = x^2 - 4, y = x^2 + 2$

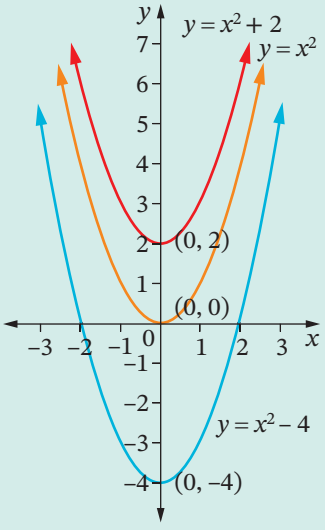
b $y = -x^2, y = -x^2 - 4, y = -x^2 + 5$

Solution

a First draw the graph of $y = x^2$. Its vertex is at (0, 0).

The graph of $y = x^2 - 4$ is like that of $y = x^2$, but moved 4 units down. Its vertex is at (0, −4).

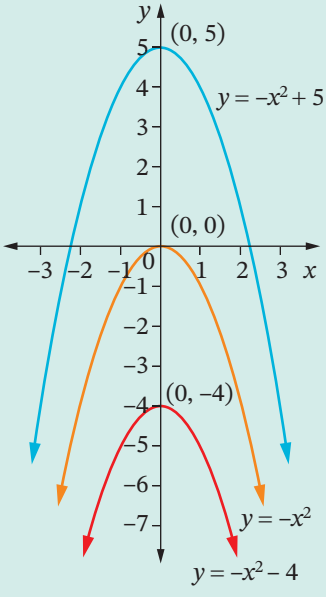
The graph of $y = x^2 + 2$ is like that of $y = x^2$, but moved 2 units up. Its vertex is at (0, 2).



b The graph of $y = -x^2$ is the graph of $y = x^2$ reflected across the x-axis. Its vertex is at (0, 0) as well.

The graph of $y = -x^2 - 4$ is like that of $y = -x^2$, but it is moved 4 units down. Its vertex is at (0, −4).

The graph of $y = -x^2 + 5$ is like that of $y = -x^2$, but it is moved 5 units up. Its vertex is at (0, 5).



Note:

- In part **a**, all parabolas are concave up, because a is positive (1)
- In part **b**, all parabolas are concave down, because a is negative (–1)
- For $y = ax^2 + c$, the y-intercept of the parabola is c

Example 8

For the graph of each quadratic equation, state:

i whether the parabola is wider or narrower than the graph of $y = x^2$

ii whether the parabola has moved up or down when compared to the graph of $y = x^2$

iii the y-intercept.

a $y = 3x^2 - 1$ **b** $y = \frac{1}{3}x^2 + 2$

Solution

a i The coefficient of x^2 is 3, while the coefficient of x^2 in $y = x^2$ is 1.

$\therefore$ The parabola will be narrower than $y = x^2$.

ii The constant term is -1.

$\therefore$ The parabola has moved down.

iii The y-intercept is -1.

Vertex at $(0, -1)$

b i The coefficient of x^2 is $\frac{1}{3}$.

$\therefore$ The parabola will be wider than $y = x^2$.

ii The constant term is 2.

$\therefore$ The parabola has moved up.

iii The y-intercept is 2.

Vertex at $(0, 2)$

Example 9

A parabola has the equation $y = 2x^2 + 3$.

Find the x-coordinate of the point on the parabola that has a y-coordinate of:

a 11 **b** 165.

Solution

a Substitute $y = 11$ into $y = 2x^2 + 3$.

$11 = 2x^2 + 3$

$8 = 2x^2$

$2x^2 = 8$

$x^2 = \frac{8}{2}$

$= 4$

$x = \pm\sqrt{4}$

$= \pm 2$

This means that there are 2 points on the parabola with a y-coordinate of 11.

They are $(2, 11)$ and $(-2, 11)$.

b Substitute $y = 165$ into $y = 2x^2 + 3$

$165 = 2x^2 + 3$

$162 = 2x^2$

$2x^2 = 162$

$x^2 = \frac{162}{2}$

$= 81$

$x = \pm\sqrt{81}$

$= \pm 9$

This makes sense because a parabola is symmetrical: (2, 11) and (–2, 11) are on either side of the y-axis.

EXERCISE 7.04 ANSWERS ON P. 599

Graphing quadratic equations $y = ax^2 + c$ U F R C

This exercise can be completed using graphing technology.

EXAMPLE 7

1 **a** Graph each quadratic equation, showing the vertex of each parabola.

$y = x^2$ $y = -x^2$ $y = x^2 + 2$ $y = -2x^2$ $y = x^2 - 1$

b State which graphs you have drawn in part **a**: C

i are concave up **ii** are concave down **iii** have a turning point at (0, 0).

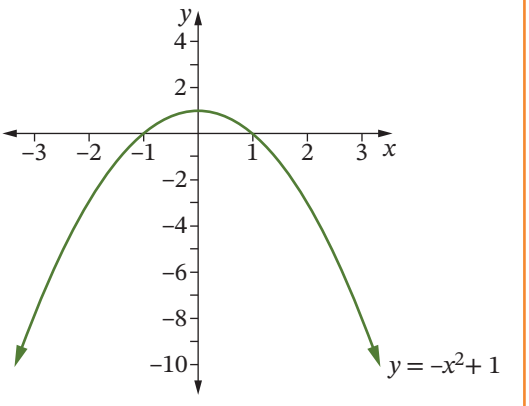
2 Which statement is false about this parabola? Select **A**, **B**, **C** or **D**. C

A Its axis of symmetry is the x-axis.

B It is concave down.

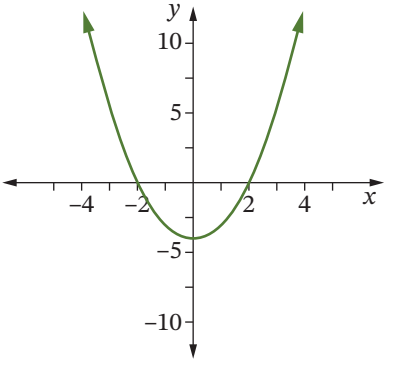
C Its vertex is (0, 1).

D It has a maximum value.

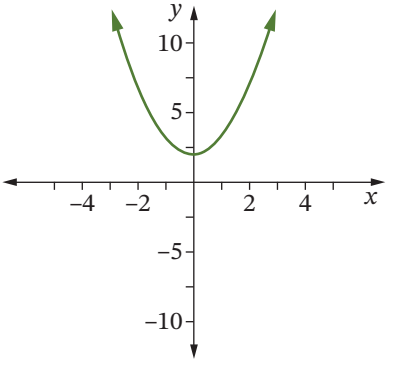


3 Which diagram shows the graph of $y = x^2 - 2$? Select **A**, **B**, **C** or **D**. R

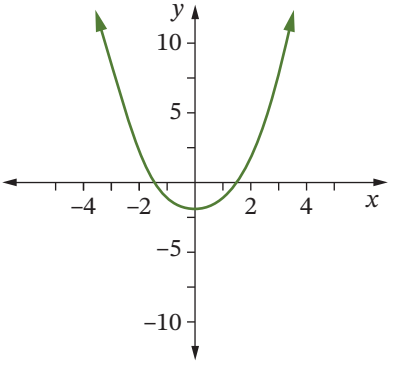
A



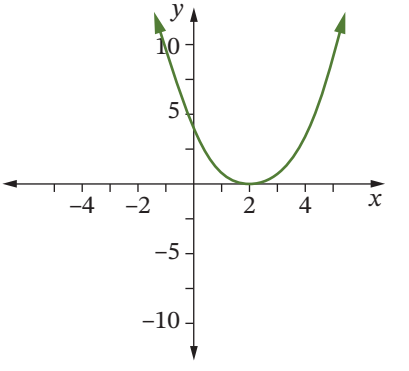
B



C



D

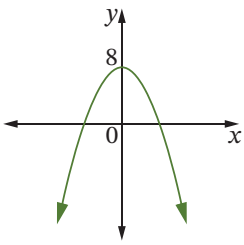


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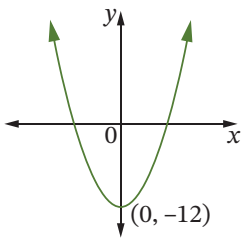
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4 Match each graph with its correct quadratic equation. R C

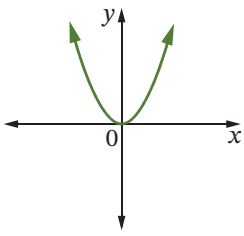
a



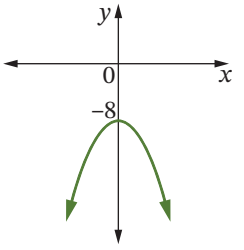
b



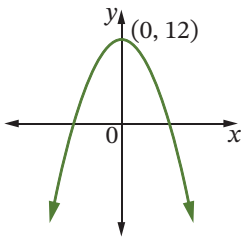
c



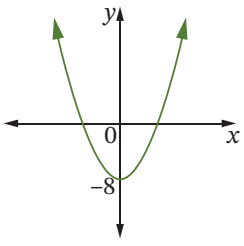
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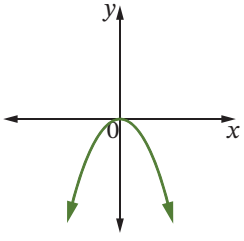
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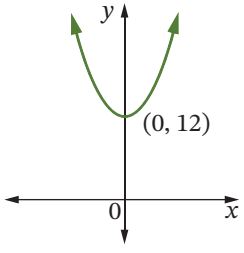
f



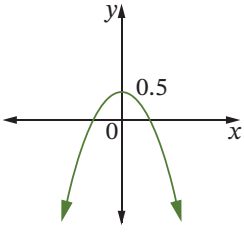
g



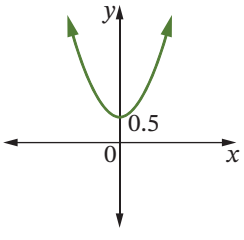
h



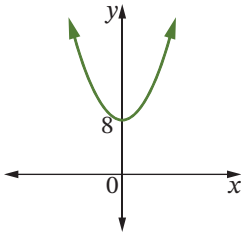
i



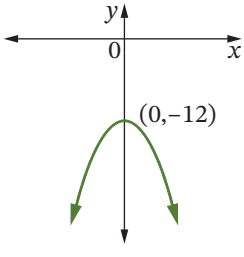
j



k



l



A $y = x^2$ **B** $y = -x^2$ **C** $y = x^2 - 8$ **D** $y = -12 - x^2$

E $y = \frac{1}{2} + x^2$ **F** $y = 8 - x^2$ **G** $y = 8 + x^2$ **H** $y = -x^2 + \frac{1}{2}$

I $y = x^2 - 12$ **J** $y = 12 - x^2$ **K** $y = -x^2 - 8$ **L** $y = x^2 + 12$

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5 Find the equation of each parabola described, in the form $y = x^2 + c$ or $y = -x^2 + c$ (where c is a constant). R C

a vertex $(0, 0)$, concave down

b concave up, turning point $(0, 0)$

c axis of symmetry $x = 0$, maximum $y = \frac{1}{4}$

d concave down, maximum $y = -9$

e turning point $\left(0, \frac{1}{2}\right)$, concave down

f axis of symmetry y-axis, minimum $y = 9$

6 a Graph $y = 2x^2 + 1$ after copying and completing this table. R C

x	−2	−1	0	1	2
y					

b State the coordinates of the vertex.

c Is the parabola concave up or concave down?

d What is its minimum value?

7 a Graph $y = -3x^2 + 2$ after copying and completing this table. R C

x	−2	−1	0	1	2
y					

b Find the vertex.

c Write the equation of its axis of symmetry.

d Find its maximum value.

8 Which statement is false about the graph of $y = 4x^2 - 1$? Select **A**, **B**, **C** or **D**. R C

A Its axis of symmetry is $y = 0$.

B It is concave up.

C The vertex is $(0, -1)$.

D It has a minimum value of $y = -1$.

EXAMPLE 8

9 For the graph of each given quadratic equation, state: R C

i whether the parabola is wider or narrower than the graph of $y = x^2$

ii whether the parabola has moved up or down when compared to the graph of $y = x^2$

iii the y-intercept.

a $y = 2x^2 + 3$

b $y = \frac{1}{2}x^2 + 1$

c $y = 6x^2 - 5$

d $y = 0.2x^2 - 12$

EXAMPLE 9

10 A parabola has the equation $y = x^2 - 5$. Find the x-coordinates for the points on the parabola that have a y-coordinate of:

a 11

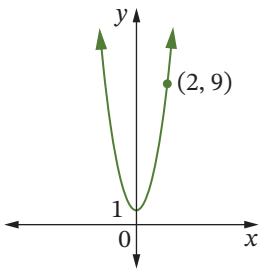
b 116.

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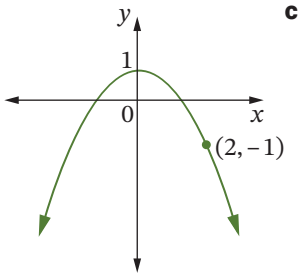
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11 Match each graph with its correct quadratic equation. R C

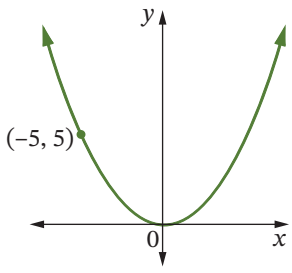
a



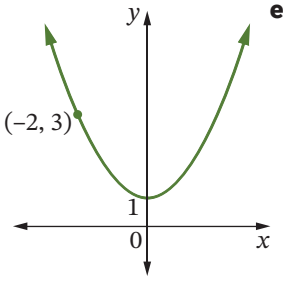
b



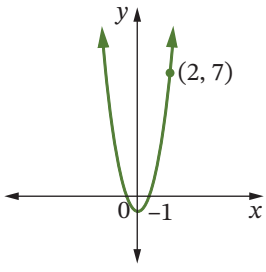
c



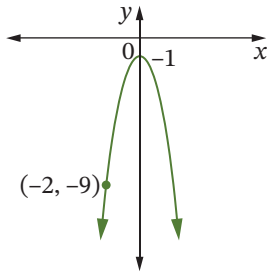
d



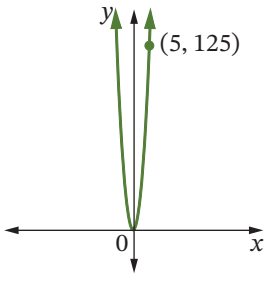
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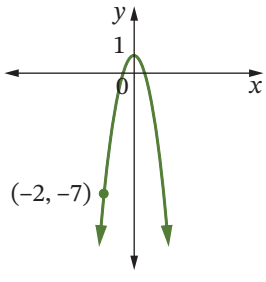
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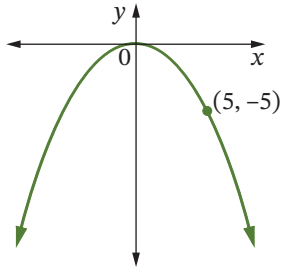
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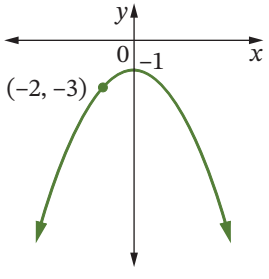
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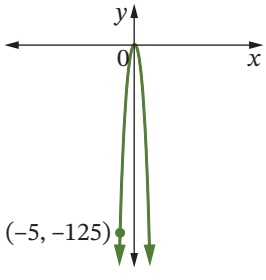
i



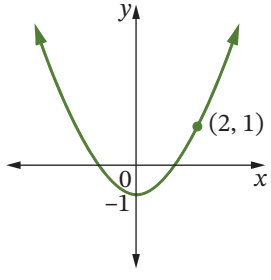
j



k



l



A $y = 5x^2$ **B** $y = 2x^2 + 1$ **C** $y = \frac{1}{2}x^2 - 1$ **D** $y = \frac{1}{5}x^2$

E $y = 2x^2 - 1$ **F** $y = -5x^2$ **G** $y = -\frac{1}{2}x^2 + 1$ **H** $y = -\frac{1}{5}x^2$

I $y = -2x^2 - 1$ **J** $y = \frac{1}{2}x^2 + 1$ **K** $y = -2x^2 + 1$ **L** $y = -\frac{1}{2}x^2 - 1$

12 A parabola has the equation $y = 2x^2 + 3$. Find the x-coordinates of the points on the parabola that have a y-coordinate of:

a 165 **b** 395.

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Did you know?

Parabolas in architecture

There are many examples of parabolas in architecture and engineering.

The Notre Dame Cathedral in Paris, France is almost 900 years old and has flying buttresses on the outside that have the shape of parabolas.



Bill Perry/Shutterstock.com

Bridges also often use parabolic curves in their construction. One modern application is the cables used in the suspension of the Golden Gate Bridge in San Francisco.



iStock.com/MasterLu

Find 2 different uses of parabolas in real-life constructions and create a presentation involving them.

Mental skills 7: Maths without calculators ANSWERS ON P. 599

Quiz
Mental skills 7

Multiplying decimals

1 Study each example.

a $3 \times 8 = 24$, so $3 \times 0.8 = 2.4$

0 dp + 1 dp = 1 dp (dp = decimal places)

The number of decimal places in the answer is equal to the total number of decimal places in the question. Also, the answer sounds reasonable because, by estimation:

$3 \times 0.8 \approx 3 \times 1 = 3$ $(2.4 \approx 3)$

b $6 \times 5 = 30$, so $0.6 \times 0.5 = 0.30 = 0.30$

1 dp + 1 dp = 2 dp

By estimation, $0.6 \times 0.5 \approx 0.5 \times 0.5 = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = 0.25$ $(0.3 \approx 0.25)$

c By estimation, $0.07 \times 0.3 \approx 0.07 \times \frac{1}{3} \approx 0.02$ $(0.021 \approx 0.02)$

2 Now evaluate each product.

a	0.7×5	**b**	12×0.2	**c**	0.4×0.3	**d**	$(0.6)^2$
e	8×0.1	**f**	0.03×0.9	**g**	4×0.05	**h**	1.1×8
i	0.3×0.8	**j**	0.2×0.06	**k**	9×0.2	**l**	0.07×0.4

3 Study each example.

Given that $15 \times 23 = 345$, evaluate each product.

a $1.5 \times 2.3 = 3.45$

1 dp + 1 dp = 2 dp (*Estimate* $1.5 \times 2.3 \approx 2 \times 2 = 4$)

b $150 \times 0.23 = 15 \times 10 \times 0.23 = 15 \times 0.23 \times 10 = 3.45 \times 10 = 34.5$

0 dp + 2 dp = 2 dp

(*Estimate* $150 \times 0.23 \approx 150 \times 0.2 = 150 \times \frac{1}{5} = 30$)

c $0.15 \times 2300 = 0.15 \times 23 \times 100 = 3.45 \times 100 = 345$

2 dp + 0 dp = 2 dp

(*Estimate* $0.15 \times 2300 \approx 0.2 \times 2300 = \frac{1}{5} \times 2300 = 460$)

4 Now given that $39 \times 17 = 663$, evaluate each product.

a	3.9×17	**b**	39×170	**c**	39×0.17	**d**	0.39×1.7
e	3.9×1.7	**f**	390×1.7	**g**	3.9×0.17	**h**	3.9×170
i	3900×1.7	**j**	39×1.7	**k**	39×0.017	**l**	0.39×0.17

7.05 Applying quadratic equations

Video
Applying quadratic functions

Parabolas can be seen and are used in our everyday lives. When water is sprayed upwards from a fountain or hose, or when a ball is thrown or kicked, the path of the object is a parabola.

Example 10

A ball is thrown and its height (h metres) at any time (t seconds) is given by the quadratic equation $h = 2t - 0.5t^2$.

Quadratic equations can have different variables other than x and y.

a Complete this table of values to graph the equation.

Time, t (seconds)	0	1	2	3	4	5
Height, h (metres)						

b What is the maximum height of the ball?

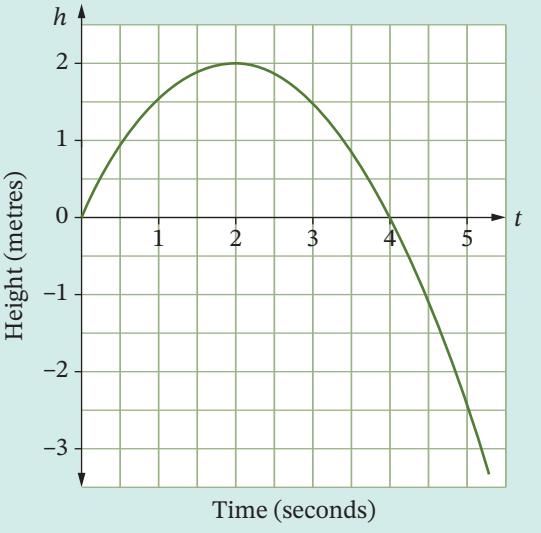
c Find the total time the ball is in the air.

d Use the graph to find when the height of the ball is 1.5 m.

Solution

a

Time, t (seconds)	0	1	2	3	4	5
Height, h (metres)	0	1.5	2	1.5	0	−2.5



Note: This is not the graph of the path of the ball. It is the graph of the height of the ball at different times, t seconds. We can ignore that part of the graph after $t = 4$ below the t-axis because the height cannot be negative.

b Maximum height is 2 m. From table or graph, $h = 2$

c 4 s. When $h = 0$ again, at $t = 4$

d The ball is at a height of 1.5 m at 1 and 3 s.

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EXERCISE 7.05 ANSWERS ON P. 600

Applying quadratic equations U F R C

EXAMPLE 10

7.05

1 A ball is thrown upwards and its height, h metres, after t seconds is given by the quadratic equation $h = 3.5t - 0.49t^2$. R C

a Copy and complete this table of values to graph the equation.

Time, t (seconds)	0	1	2	3	4	5	6	7	8
Height, h (metres)									

b What is the maximum height of the ball and when does it reach this height?

c At what times does the ball reach a height of 3 m? Explain why there are 2 answers.

2 The shape of cables hanging between 2 towers 100 m apart on a suspension bridge is modelled by the equation $h = 10 + 0.004x^2$ where h is the height, in metres above ground and x is the horizontal distance in metres from the centre of the bridge. R C

a Copy and complete this table, then graph the equation.

x	−50	−40	−30	−20	−10	0	10	20	30	40	50
h											

b How tall are the towers at either end of the cables?

c What is the height of the cables at their lowest point?

d How far from the centre is the height of the cables 15 m?

3 A stone is dropped from a cliff and its height (h metres) at any time (t seconds) is given by $h = 80 - 4.9t^2$. R C

a Graph the quadratic equation for values of t from 0 to 5.

b What is the height of the cliff?

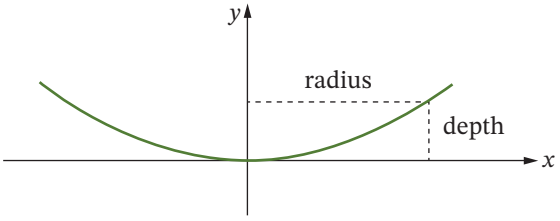
c Use your graph to find the height of the stone after 3 seconds.

d When will the stone hit the ground?

e How long after it is dropped is the stone 5 metres above the ground? Answer to the nearest 0.1 s.

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4 The inside of a large radio telescope is the shape of a parabola and can be modelled by the equation $y = 0.0091x^2$, where y metres is the depth and x metres is the radius. R



a Copy and complete this table of values.

Radius, x (metres)	−40	−30	−20	−10	0	10	20	30	40
Depth, y (metres)									

b Graph the equation, using a scale of 1 cm = 10 m on the x-axis and 1 cm = 5 m on the y-axis.

c If the radius of the radio telescope is 32 m, use your graph to find its depth.

d Find the radius of the radio telescope that has a depth of:

i 1 m **ii** 2.5 m

5 A ball is dropped from the top of a building 200 m high and the distance, d metres, it falls after t seconds is given by the equation $d = 5t^2$. R

a Copy and complete this table of values and graph the equation.

Time, t (seconds)	0	1	2	3	4	5	6	7
Distance, d (metres)								

b How far has the ball dropped after:

i 2 seconds? **ii** 5 seconds?

c How far did the ball drop in:

i the 3rd second? **ii** the 4th second?

d After how many seconds did the ball hit the ground (correct to 1 decimal place)?

e After how many seconds was the ball 80 m *above the ground*?

6 A parabolic reflector has the equation $y = 0.2x^2$, where x (cm) is the radius of the reflector and y (cm) is the depth.

a Copy and complete this table and graph the equation.

Radius, x (cm)	−5	−4	−3	−2	−1	0	1	2	3	4	5
Depth, y (cm)											

b If the radius of a reflector is 2.5 cm, what is its depth?

c If the depth of a reflector is 4.5 cm, what is its radius?

7 At a water fountain, the height of the water being sprayed up follows the equation $h = 1.25 + 4t - 0.85t^2$, where h (metres) is the height of the water and t (seconds) is the time. R C



iStock.com/Petra Richli

a Copy and complete this table, then graph this equation.

Time, t (seconds)	0	1	2	3	4	5	6
Height, h (metres)							

b What is the vertical intercept of the graph and what does this value represent?

c How long does it take for the water to leave the fountain to when it hits the ground?

d What is the maximum height the water reaches?

e How long does it take for the water to reach its maximum height?

8 The height h metres above ground of a skydiver jumping from a plane is given by $h = 3000 - 5t^2$, where t seconds is the time after jumping.

a Copy and complete this table to graph the equation. Use a scale of 1 cm = 5 s on the horizontal axis and 1 cm = 500 m on the vertical axis.

Time (t seconds)	0	5	10	15	20	25
Height (y metres)						

b What was the height of the plane when the skydiver jumped?

c What was the height of the skydiver after 15 seconds?

d If the parachute is opened at a height of 1200 m, how long will it take the skydiver to drop to this height?

7.05

Investigation

PATH

Graphing $y = \frac{1}{x}$

1 Copy and complete this table for $y = \frac{1}{x}$. Explain why no y value exists for $x = 0$.

x	−5	−4	−3	−2	−1	−0.5	−0.2	−0.1	0	0.1	0.2	0.5	1	2	3	4	5
y									–								

2 Hence graph $y = \frac{1}{x}$ on a number plane.

3 There are 2 parts or 'branches' to your graph. In which quadrants of the number plane are the branches?

4 Use your graph to explain what happens to the y value as x becomes very large.

5 Explain what happens to the y value as x approaches 0.

6 The graph of $y = \frac{1}{x}$ has 2 axes of symmetry. Draw them on your graph.

7 Copy and complete the table from question **1** for $y = -\frac{1}{x}$.

8 Hence graph $y = -\frac{1}{x}$ on a number plane.

9 How does the graph of $y = -\frac{1}{x}$ compare with that of $y = -\frac{1}{x}$?

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Technology

Graphing $y = \frac{k}{x}$

1 Use graphing technology to graph each equation.

a $y = \frac{1}{x}$ **b** $y = \frac{2}{x}$ **c** $y = \frac{5}{x}$ **d** $y = \frac{10}{x}$

2 Compare the graphs from question **1**. What happens to the graph of $y = \frac{k}{x}$ as k increases?

3 Graph $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ and compare them.

4 Graph $y = \frac{4}{x}$ and use **Trace** to complete this table of values.

x	1	2	5	10	100	200	1000
y							

5 What happens to the y values when the x values become very large?

6 For $y = \frac{4}{x}$ use the **Trace** function to complete this table of values.

x	0.0001	0.01	0.1	0.5	1	5
y						

7 What happens to the y values when the x values become very small and close to 0?

7.06 Graphing hyperbolas $y = \frac{k}{x} + c$

PATH

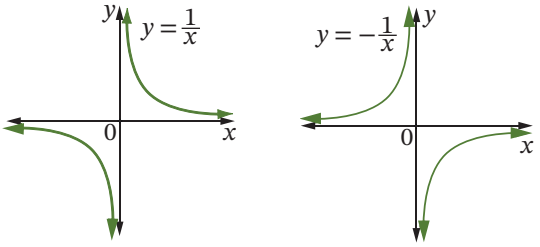
Worksheet
Graphing hyperbolas

Technology
Curve sketcher

Spreadsheet
Curve sketcher

The graph of $y = \frac{k}{x}$, where k is a constant, is a curve with 2 branches called a **hyperbola** (pronounced 'hy-perb-o-la').

The graphs of $y = \frac{1}{x}$ and $y = -\frac{1}{x}$ are shown.



The hyperbola $y = \frac{k}{x}$

- The hyperbola has 2 separate branches in different quadrants.
- If k is positive, the graph is in the 1st and 3rd quadrants.
- If k is negative, the graph is in the 2nd and 4th quadrants.
- The graph has 2 axes of symmetry: their equations are the straight lines $y = x$ and $y = -x$.
- The graph has rotational symmetry of 180° about (0, 0).
- The higher the value of k, the further the hyperbola is from the x- and y-axes.
- As x becomes very large, y gets closer to 0.
- As x becomes closer to 0, y gets very large.
- The graph gets very close to the x- and y-axes but never crosses them. The x- and y-axes are called **asymptotes** because the graph approaches them but never touches them.

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Example 11

Graph each hyperbola and mark the coordinates of one point on the curve.

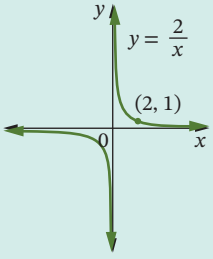
a $y = \frac{2}{x}$ **b** $y = -\frac{3}{x}$

Solution

a Let $x = 2$ Choosing any value of x

$$y = \frac{2}{2}$$
$$= 1$$

A point on the curve is (2, 1).

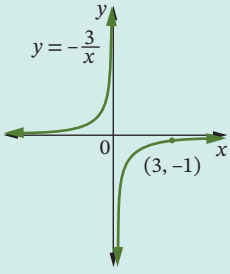


b Let $x = 3$

$$y = \frac{3}{3}$$
$$= -1$$

A point on the curve is (3, –1).

As $k = -3$ is negative, the hyperbola is in the 2nd and 4th quadrants.



The hyperbola $y = \frac{k}{x} + c$

The graph of $y = \frac{k}{x} + c$ is the hyperbola $y = \frac{k}{x}$ translated c units up (or down if c is negative).

The hyperbola $y = \frac{k}{x - b}$

The graph of $y = \frac{k}{x-b}$ is the hyperbola $y = \frac{k}{x}$ translated b units to the right (or left if b is negative).

Example 12

Graph each hyperbola, find any intercepts and mark the coordinates of one point on the curve.

a $y = \frac{2}{x} + 1$ **b** $y = -\frac{3}{x-2}$

Solution

a Let $x = 2$ Choosing any value of x

$$y = \frac{2}{2} + 1$$
$$= 2$$

A point on the curve is (2, 2).

PATH

Video
Graphing hyperbolas

7.06

PATH

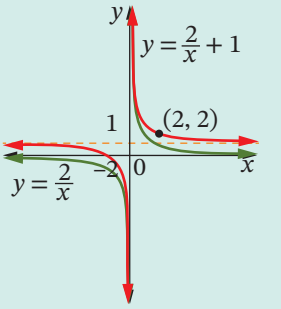
The graph of $y = \frac{2}{x} + 1$ is the graph of $y = \frac{2}{x}$ translated up 1 unit. This means that the horizontal asymptote is now at $y = 1$.
An x-intercept now occurs when $y = 0$.

$0 = \frac{2}{x} + 1$ Multiplying both sides by x

$0 = 2 + x$

$x = -2$

The x-intercept is -2.



b Let $x = 3$

$y = -\frac{3}{3-2}$

$= -3$

A point on the curve is $(3, -3)$.

The graph of $y = -\frac{3}{x-2}$ is the graph of $y = -\frac{3}{x}$ translated right 2 units. This means that the vertical asymptote is now at $x = 2$.

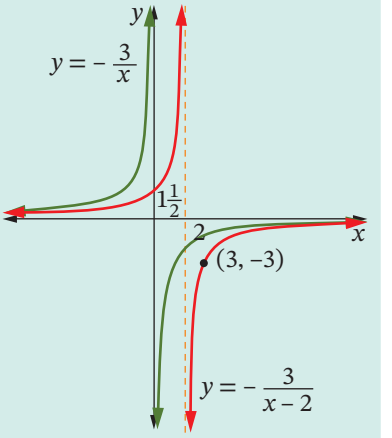
A y-intercept now occurs when $x = 0$.

$y = -\frac{3}{0-2}$

$= \frac{-3}{-2}$

$= 1\frac{1}{2}$

The y-intercept is $1\frac{1}{2}$.



EXERCISE 7.06 ANSWERS ON P. 601

Graphing hyperbolas $y = \frac{k}{x} + c$ UFRC

1 a Copy and complete this table for $y = \frac{2}{x}$.

x	−3	−2	−1	0	1	2	3
y				–			

b Graph $y = \frac{2}{x}$, showing the coordinates of one point on the hyperbola.

c On your diagram, draw in the axes of symmetry for the hyperbola.

d What are the equations of these axes?

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EXAMPLE 11

7.06

2 Graph each hyperbola and mark the coordinates of one point on the curve.

a $y = \frac{4}{x}$ **b** $y = -\frac{2}{x}$ **c** $y = \frac{3}{x}$

3 **a** The distance from Sydney to Melbourne is close to 1000 km. Using the equation $s = \frac{1000}{t}$, copy and complete the following table that relates time (t hours) and speed (s km/h) for the trip. Round your answers to the nearest km/h.

t	1	2	3	...	10
s					

b Hence graph the equation $s = \frac{1000}{t}$.

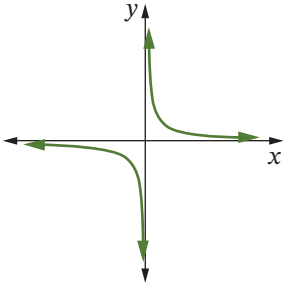
c Why are the values for t only positive numbers? Explain why t cannot be equal to 0. R C

d If the time is doubled, is the speed halved? Use the information from your graph to support your answer. R C

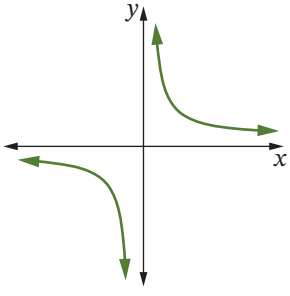
4 Match each equation with its graph. R

a $y = \frac{5}{x}$ **b** $y = -\frac{2}{x}$ **c** $y = \frac{1}{x}$ **d** $y = -\frac{5}{x}$

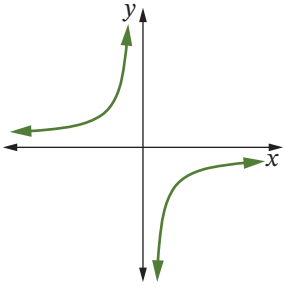
A



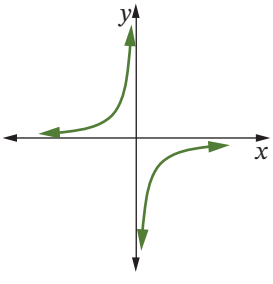
B



C



D



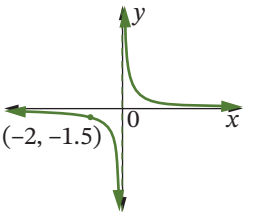
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5 This curve is a hyperbola of the form $y = \frac{k}{x}$.

a Find the value of k.

b Hence state the equation of this hyperbola.



EXAMPLE 12

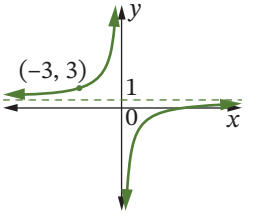
6 Graph each hyperbola and mark the coordinates of one point on the curve.

a $y = \frac{1}{x} + 2$ **b** $y = -\frac{2}{x} - 3$ **c** $y = \frac{2}{x-1}$ **d** $y = \frac{-3}{x+2}$

7 This curve is a hyperbola with equation $y = \frac{k}{x} + c$.

a Find the values of c and k.

b Hence state the equation of this hyperbola.



8 Sarah and David want to buy a rectangular block of land that has an area of 800 m^2. There are several blocks available with this area. R C

a Copy and complete this table that relates the length (L metres) and width (W metres) of the block of land.

L	10	20	30	...	100
W					

b What is the formula for W?

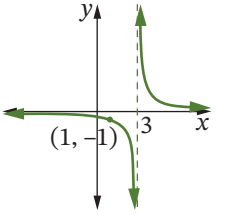
c Explain why the length or width cannot be equal to 0 metres.

d Graph the formula for W.

e What happens to the width as the length continues to increase? How is this shown on the graph?

f What happens to the width as the length approaches 0? How is this shown on the graph?

9 Which equation best represents this hyperbola?
Select the correct answer **A**, **B**, **C** or **D**.



A $y = \frac{2}{x-3}$ **B** $y = \frac{-1}{x+3}$

C $y = \frac{2}{x} + 3$ **D** $y = -\frac{1}{x} - 3$

10 Find the asymptotes of the hyperbola with equation: R C

a $y = \frac{k}{x}$ **b** $y = \frac{k}{x} + c$ **c** $y = \frac{k}{x-b}$ **d** $y = \frac{k}{x-b} + c$

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Investigation

Graphing $y = 2^x$

This activity can be completed using graphing technology.

1 Copy and complete this table of values for $y = 2^x$.

x	−3	−2	−1	0	1	2	3	4
y								

2 Graph the points from the table and join them with a smooth curve. The equation $y = 2^x$ is called an exponential equation and its graph is called an exponential curve (exponent means 'power').

3 Graph $y = 2^{-x}$ in a similar way.

4 Compare the graphs of $y = 2^x$ and $y = 2^{-x}$. Describe any similarities and differences.

5 The y-intercept of any graph with equation $y = a^x$ (where a is a positive constant) is always 1. Explain why.

6 The graph of $y = 2^x$ is increasing. Is the graph of $y = 2^{-x}$ increasing or decreasing? Give reasons.

7 Describe what happens to the graph of $y = 2^x$ when:

a x approaches a large positive number

b x approaches a large negative number.

Technology

Exponential curves

Use graphing technology to complete this activity.

1 Graph $y = 2^x$ using the software. Adjust the colour or line thickness if you need to.

2 Now graph these exponential curves, using different colours.

$y = 2^{-x}$, $y = -2^x$, $y = -2^{-x}$, $y = 2^x + 1$, $y = 2^x - 1$

3 **a** Which graphs are similar?

b Identify any features such as y-intercepts.

c Which graphs are similar as:

i x becomes larger? **ii** x becomes smaller?

4 Repeat steps **1** to **3** for these exponential curves:

$y = 3^x$, $y = 3^{-x}$, $y = -3^x$, $y = -3^{-x}$, $y = 3^x + 1$, $y = 3^x - 1$

7.07 Graphing exponential equations $y = a^x + c$

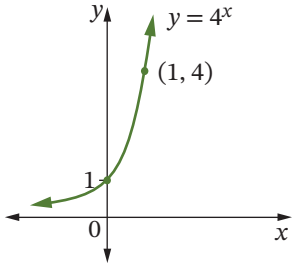
Videos
The exponential curve
The emperor's chessboard

Worksheet
Graphing exponentials

An equation of the form $y = a^x$, where a is a positive constant and the variable x is a power, is called an **exponential equation**, for example, $y = 5^x$, $y = 2^x$ and $y = 3^x$. The graph of an exponential equation is called an **exponential curve**.

The table of values and graph of $y = 4^x$ is shown.

x	−2	−1	0	1	2	3	4
y	$\frac{1}{16}$	$\frac{1}{4}$	1	4	16	64	256



- The y-intercept of $y = a^x$ is always 1 since $a^0 = 1$.
- As x increases (to the right, in the positive direction), a^x becomes very large. Graphically, this means that the graph of $y = a^x$ increases sharply with a steep positive gradient.
- As x decreases (to the left, in the negative direction), a^x approaches 0. This means that the graph of $y = a^x$ flattens out and approaches the x-axis as x becomes a large negative number. The x-axis is an **asymptote** because the curve approaches it but never touches it.
- The exponential curve is always above the x-axis because the value of a^x is always positive.

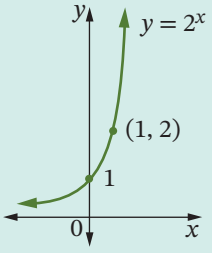
Example 13

Sketch each exponential equation and mark the y-intercept on each curve.

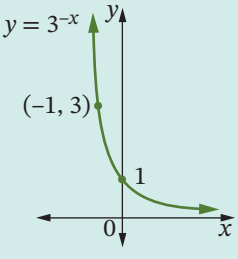
a $y = 2^x$ **b** $y = 3^{-x}$

Solution

a
- The y-intercept of $y = 2^x$ is 1
- When $x = 1$, $y = 2$, so (1, 2) is a point on the curve
- As x increases (to the right in the positive direction), 2^x becomes very large (steep positive gradient)
- As x decreases (to the left in the negative direction), 2^x approaches 0
- The x-axis is an asymptote.



b
- The y-intercept of $y = 3^{-x}$ is 1
- When $x = -1$, $y = 3$, so (−1, 3) is a point on the curve
- As x decreases (to the left in the negative direction), 3^{-x} becomes very large (steep negative gradient)
- As x increases (to the right in the positive direction), 3^{-x} approaches 0
- The x-axis is an asymptote.



Note that the graph of $y = 3^{-x}$ (and of $y = a^{-x}$ in general) is decreasing, and is actually a reflection of $y = 3^x$ in the y-axis.

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EXERCISE 7.07 ANSWERS ON P. 603

Graphing exponential equations $y = a^x + c$ U F R C

Some of this exercise may be completed using graphing technology.

EXAMPLE 13

7.07

1 **a** Graph each exponential equation on the same axes. R C

i $y = 2^x$ **ii** $y = 3^x$ **iii** $y = 5^x$

b What is the y-intercept of each curve?

c Describe what happens to the graph of $y = a^x$ as a increases.

2 **a** Graph $y = 4^x$ and $y = 4^{-x}$ on the same axes.

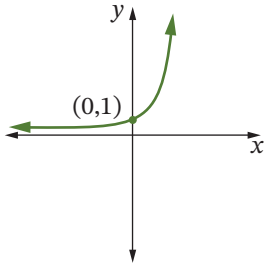
b Copy and complete:

i The reflection of $y = 4^x$ in the y-axis is ...

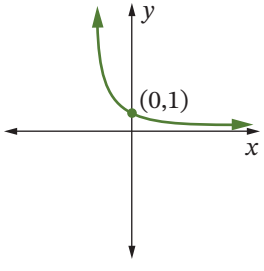
ii The reflection of $y = a^x$ in the y-axis is ...

3 Which graph represents $y = 2^{-x}$? Select the correct answer **A**, **B**, **C** or **D**. R C

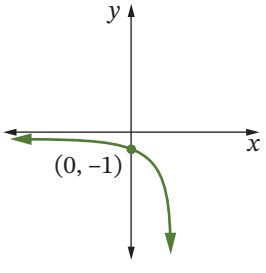
A



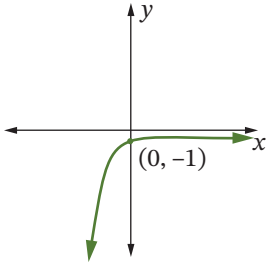
B



C



D



4 **a** Graph $y = 2^x$ and $y = -2^x$ on the same axes.

b How are the 2 graphs related? R C

c Copy and complete: The reflection of $y = a^x$ in the x-axis is ...

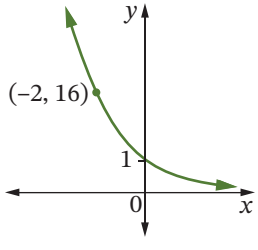
5 Graph $y = 3^x + 1$ and $y = 3^x - 1$ on the same axes and describe how they are related. R C

6 Sketch each exponential curve, showing the y-intercept.

a $y = 2^x$ **b** $y = 3^{-x}$ **c** $y = -4^x$

d $y = -2^{-x}$ **e** $y = 4^x + 1$ **f** $y = 4^x - 1$

7 Find an exponential equation for this graph. R C



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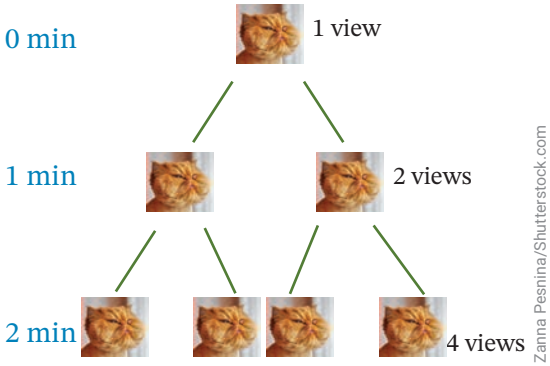
Investigation

Videos going viral

Use a spreadsheet to help you with this investigation.

With the invention of social media and social networks, we can now communicate with each other instantly. However, what often starts off as a simple message between friends or an online video post can quickly multiply at an alarming rate, be seen by the general public worldwide and 'go viral' (spread like a virus).

Suppose a video of an angry cat is shared with a friend who one minute later shares it with 2 other friends. This occurs every minute so that every minute the number of *new* views doubles. This is an example of something growing exponentially, according to the formula $y = 2^x$, where x represents number of minutes.



Zanna Pesnina/Shutterstock.com

1 Use a spreadsheet to calculate the number of new views after:

a 15 min **b** 30 min **c** 1 hour **d** 2 hours

e 4 hours **f** 10 hours **g** 20 hours **h** 30 hours

2 Find how long it will take until the total number of new views reaches:

a 64 **b** 256 **c** over 500 **d** over 1000

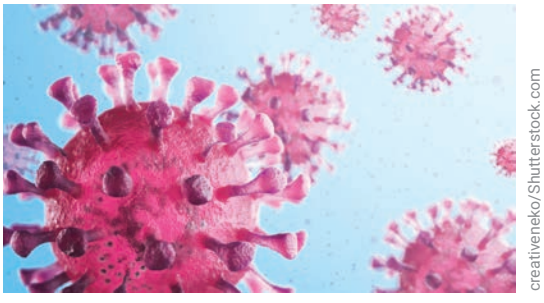
e over 3000 **f** over 10 000 **g** over 1 million **h** over 4 million

3 Use the spreadsheet's **Graph Option** to graph the viral pattern. Describe the shape of the graph.

Did you know?

Exponential growth

When an increase can be described using an exponential equation, it is called exponential growth. Examples include the growth of a virus, population (people or animals) and compound interest investments.


creativeneko/Shutterstock.com

Population growth is monitored in different countries through the fertility (birth) and mortality (death) rates as well as migration. The data collected for these figures can often be modelled as an exponential function. By modelling the changes in population, predictions of future changes in population can be simulated and towns and cities can prepare for possible expansion in the numbers of schools, hospitals, housing and other necessary infrastructure.

At what rate is the population of Australia growing?
What about the world's population?

Investigation

Graphing $y = 3(2^x)$

This activity can be completed using graphing technology.

1 Copy and complete this table of values for $y = 3(2^x)$.

$3(2^x)$ means $3 \times (2^x)$

x	−3	−2	−1	0	1	2	3	4
y								

2 Graph the points from the table and join them with a smooth curve.

3 Graph each exponential equation in a similar way.

a $y = 1.5(2^x)$ **b** $y = 4(3^x)$ **c** $y = 0.5(4^x)$

4 Compare the graphs $y = 3(2^x)$, $y = 1.5(2^x)$, $y = 4(3^x)$ and $y = 0.5(4^x)$. Describe any similarities and differences.

5 Graph each exponential equation and describe any similarities and differences.

a $y = 5(2^{-x})$ **b** $y = 4(3^{-x})$ **c** $y = 0.5(4^{-x})$ **d** $y = 2(5^{-x})$

6 **a** What are the y-intercepts of the graphs in Question 1?

b What are the y-intercepts of the graphs in Question 2?

c What is the y-intercept of any graph with equation $y = k(a^x)$ or $y = k(a^{-x})$?

Exponential growth and decay 7.08

Any quantity that increases according to the exponential equation $y = k(a^x)$, where $a > 1$, is showing **exponential growth**, increasing slowly at first and then more quickly.

Examples of exponential growth are population growth, an investment increasing with compound interest, and the number of people or computers affected by a virus.

Any quantity that decreases according to the exponential equation $y = k(a^{-x})$, where $a > 1$, is showing **exponential decay**, decreasing quickly at first and then more slowly.

Examples of exponential decay are the cooling of a hot liquid, the depreciating value of an asset and radioactive material decay.

We have studied compound interest and depreciation in Chapter 1, *Interest and depreciation*.

Example 14

The future value $\$FV$ of an amount of \$80, invested at 5% p.a. interest compounded annually, after n years is given by the formula $FV = 80(1.05)^n$.

a Complete this table of values, correct to the nearest dollar.

n	0	10	20	30	40	50
FV						

b Draw the graph of the equation. Use a scale of 1 cm = 10 years on the x-axis and 1 cm = \$100 on the y-axis.

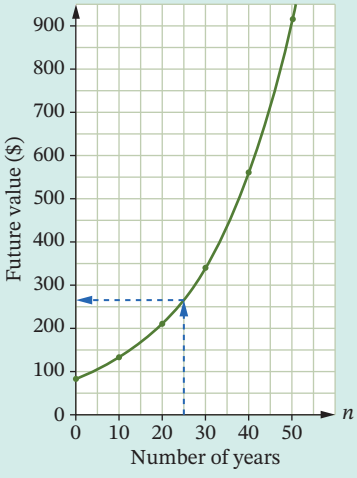
c Use your graph to find the future value of the investment after 25 years.

Solution

a

n	0	10	20	30	40	50
FV	80	130	212	346	563	917

b, c



Using the graph, the value of the investment after 25 years is about \$270.

(Using the formula, the value of the investment is \$270.91)

Example 15

The value $\$V$ of a car which depreciates annually is given by the equation $V = 50\,000\,(1.25)^{-n}$, where n is the number of years.

a Graph the equation for values of n from 0 to 5.

b What is the vertical intercept and what does it represent?

c Use the graph to estimate the value of the car after 3.5 years.

d Use the equation to determine when the car's value will drop to below \$10 000.

Solution

a Completing a table of values with V rounded to the nearest \$100, using the formula $V = 50\,000\,(1.25)^{-n}$.

n	0	1	2	3	4	5
V	50 000	40 000	32 000	25 600	20 500	16 400

b The vertical intercept is 50 000, which is the original value of the car.

c From the graph, the value of the car after 3.5 years is \$23 000.

(Using the formula, the value of V is \$22 897)

d Use guess-and-check:

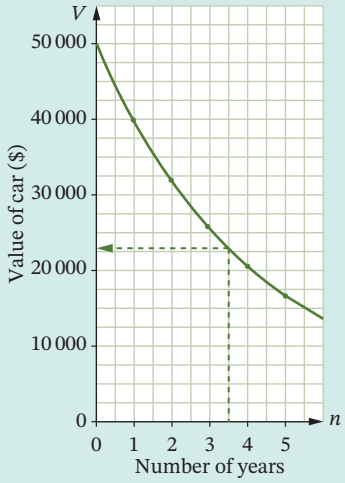
$n = 5, V = 16\,400$

$n = 6, V = 13\,107$

$n = 7, V = 10\,485$

$n = 8, V = 8388$

The value of the car will be less than \$10 000 after 8 years.



EXERCISE 7.08 ANSWERS ON P. 605

Exponential growth and decay U F R C

EXAMPLE 14

1 The population of a small town grows according to the equation $P = 1200\,(1.095)^n$, where P is the population and n is the number of years after the end of 2025. R C

a Copy and complete the table of values correct to the nearest 10.

n	0	1	2	3	4	5	6	7	8	9	10
P											

b Graph the exponential equation for n from 0 to 10. Use a scale of 1 cm = 2 years on the horizontal axis and a scale of 1 cm = 400 people on the vertical axis.

c What is the vertical intercept and what does it represent?

d Use a guess-and-check method to find in what year the population will be triple the original population.

2 Bacteria cells grow by continually dividing into 2 cells. So 1 cell will divide into 2 cells, the 2 cells divide into 4 cells and so on. The number of bacteria cells (B) present can be found using the exponential equation $B = 10(2)^n$, where n is the number of times the cells divide. The cells divide every 20 minutes. R C

a Copy and complete this table of values and then graph the equation.

Use a scale of 1 cm = 1 division on the horizontal axis and 1 cm = 100 bacteria on the vertical axis.

n	0	1	2	3	4	5	6
B							

b How many bacteria were there initially?

c How many bacteria were present after:

i 2 hours? **ii** 3 hours? **iii** 5 hours?

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3 The future value of $1500 invested at 6.5% p.a. interest compounded annually for n years is given by the equation $FV = 1500(1.065)^n$. R C

a Copy and complete this table correct to the nearest dollar.

n	0	2	4	6	8	10
FV						

b Graph the equation using a scale of 1 cm = 2 years on the horizontal axis and 1 cm = $200 on the vertical axis

c What is the vertical intercept and what does it represent?

d Use your graph to find the what the future value is after:

i 3 years **ii** 7 years

EXAMPLE 15

4 A ball is dropped from a window and the height (H cm) it reaches each bounce (x) is given by the equation $H = 400(1.55)^{-x}$. R C

a Copy and complete this table of values (to the nearest whole number).

x	0	2	4	6	8	10
H						

b Graph the equation using a scale of 1 cm = 2 bounces on the horizontal axis and 1 cm = 50 cm on the vertical axis.

c What is the vertical intercept and what does it represent?

d Use the graph to find the height of the ball on the 5th bounce.

e After which bounce will the height be less than 20 cm?

f Use the equation to find the height of the ball on the 14th bounce.

5 The value of a car that depreciates annually is given by the equation $V = 35\,000(1.111)^{-n}$, where n is the number of years. R C

a Graph the equation for values of n from 0 to 8.

b What is the vertical intercept and what does it represent?

c Use the graph to estimate the value of the car after 3.5 years.

d Use the equation to determine when the car's value will drop to below $10 000.

6 Atmospheric pressure (A), measured in millibars (mbar) decreases with altitude (h, metres) and can be calculated using the exponential equation $A = 1013\left(\frac{25}{22}\right)^{-0.001h}$. R C

a Copy and complete this table of values and draw the graph. Use a scale 1 cm = 1000 m on the horizontal axis and 1 cm = 100 mbar on the vertical axis.

h	0	1000	2000	3000	4000	5000	6000	8000	10 000
A									

b What is the vertical intercept and what does it represent?

c Use your graph to find the atmospheric pressure at:

i 2500 m **ii** 5100 m **iii** 9000 m

d Use your graph to find the altitude if the atmospheric pressure is:

i 700 mbar **ii** 500 mbar **iii** 350 mbar

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e Use the graph to find the atmospheric pressure at the summit of:

i Mt Kosciuszko (2228 m) **ii** Mt Everest (8848 m)

f Use the equation to find the atmospheric pressure at a height of 15 000 m.

7 The half -life of caffeine in the human body is about 5 hours. This means that if you consume 20 mg of caffeine, then after 5 hours you will still have 10 mg in your body.

An equation for the amount of caffeine (C mg) left in the human body t hours after drinking a cup of coffee (about 230 mL) is $C = 95(2)^{-0.2t}$. R C

a Copy and complete this table of values and graph the equation.

t	0	5	10	15	20	25
C						

b How much caffeine was originally in the cup of coffee?

c Use the graph to find how much caffeine will be in the human body after:

i 7 hours **ii** 16 hours **iii** 24 hours

d After how many hours will the amount of caffeine be less than 20 mg?

8 House prices ($P \times \$100\,000$) in a city are increasing according to the equation $P = P_0(1.15)^n$ where $P_0 \times \$100\,000$ is the original price of a house and n is the number of years after the end of 2025. Note that P and P_0 are values multiplied by \$100 000. R C

a The original price of a house is \$500 000.
Copy and complete this table, correct to one decimal place.

n	0	1	2	3	4	5
P (in \$100 000s)	5					

b Draw the graph of the equation.

c What is the vertical intercept and what does it represent?

d Use the graph to find the value of the house at the end of 2030.

9 A certain radioactive substance decays such that its mass (M kg) after t years is given by the equation $M = 100(2.718)^{-0.016t}$. R C

a Copy and complete this table of values correct to one decimal place.

t	0	10	20	30	40	60
M (kg)						

b What was the original mass of the radioactive substance?

c Draw the graph of the exponential equation. Use a scale of 1 cm = 10 years on the horizontal axis and 1 cm = 10 kg on the vertical axis.

d Use your graph to find the mass remaining after 25 years.

e The half-life of the substance is the time it takes to decay to half of the original mass. Use the graph to determine the half-life of the substance.

f Use the equation to find correct to 2 decimal places what amount of the substance remains after:

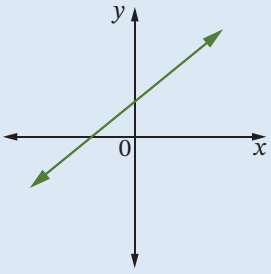
i 100 years **ii** 200 years **iii** 300 years

7.09 Identifying graphs

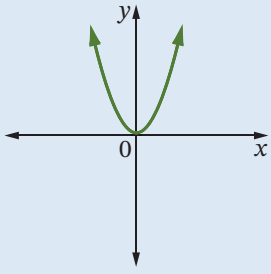
Puzzle
Matching graphs

Graphs

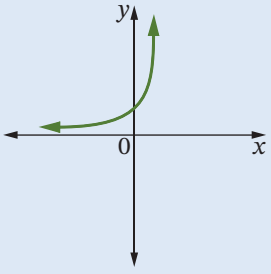
Straight line $y = mx + c$ or $ax + by + c = 0$



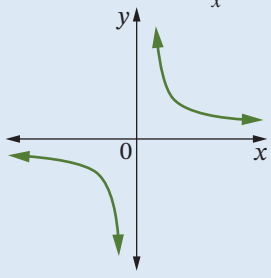
Parabola $y = ax^2 + c$



Exponential curve $y = a^x$



Hyperbola $y = \frac{k}{x}$



Note: The hyperbola $y = \frac{k}{x}$ is PATH content.

Example 16

State whether each equation represents a straight line, a parabola, a hyperbola or an exponential curve.

a $y = 3x^2 - 1$ **b** $y = 5x + 7$ **c** $y = -\frac{1}{2}x^2 - 10$

d $y = \frac{2}{x}$ **e** $y = 3^x$

Solution

a $y = 3x^2 - 1$ is a parabola because it is of the form $y = ax^2 + c$.

b $y = 5x + 7$ is a straight line because it is of the form $y = mx + c$.

c $y = -\frac{1}{2}x^2 - 10$ is a parabola because it is of the form $y = ax^2 + c$.

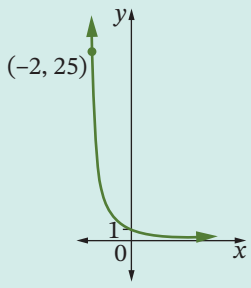
d $y = \frac{2}{x}$ is a hyperbola because it is of the form $y = \frac{k}{x}$.

e $y = 3^x$ is an exponential curve because it is of the form $y = a^x$.

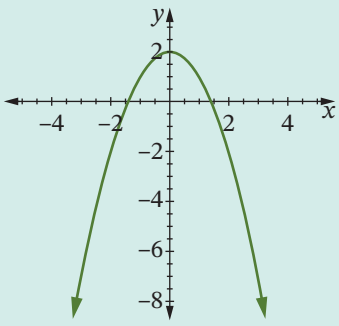
Example 17

Match each graph with its equation.

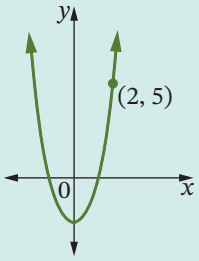
a



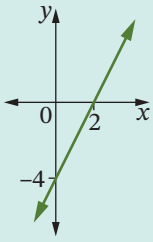
b



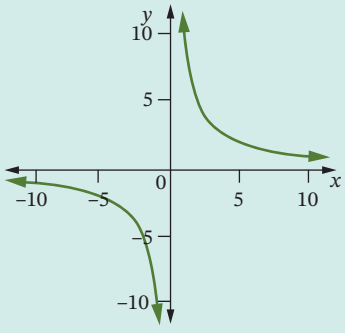
c



d



e



A $y = 2x - 4$ **B** $y = \frac{9}{x}$ **C** $y = 2x^2 - 3$

D $y = 5^{-x}$ **E** $y = -x^2 + 2$

7.09

Solution

When matching graphs to equations, the coordinates of a point on the graph may need to be substituted into the equation to verify that the equation represents the graph.

a This is an exponential curve. The only possible match is **D**, $y = 5^{-x}$

Test point: $(-2, 25)$

LHS $= 25$

RHS $= 5^{-(-2)} = 5^2 = 25 =$ LHS

b This is a concave down parabola with a y-intercept of 2. The only possible match is **E**, $y = -x^2 + 2$

c This is a concave up parabola that matches with **C**, $y = 2x^2 - 3$

Test point: $(2, 5)$

LHS $= 5$

RHS $= 2 \times 2^2 - 3 = 5 =$ LHS

d This is a straight line with a y-intercept of -4 that matches with **A**, $y = 2x - 4$

e This is a hyperbola that has points $(3, 3)$ and $(-3, -3)$.

The only possible match is **B**, $y = \frac{9}{x}$

EXERCISE 7.09 ANSWERS ON P. 607

Identifying graphs UFRC

Note: The hyperbola is PATH content.

EXAMPLE 16

1 For each equation state whether its graph is a straight line (L), a parabola (P), an exponential curve (E) or a hyperbola (H). R C

a	$y = 9x^2 - 4$	**b**	$y = 9x$	**c**	$y = 9^x$	**d**	$y = 9$
e	$y = \frac{12}{x}$	**f**	$y = 3x - 8$	**g**	$y = 3x^2 - 8$	**h**	$y = 2x + 5$
i	$y = -x^2 + 6$	**j**	$y = 10^{-x}$	**k**	$y = 7x^2 + 2$	**l**	$y = -\frac{2}{x}$

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EXAMPLE 17

7.09

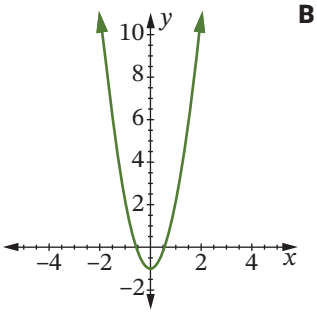
2 Match each equation with its graph. R C

a $x = 4$ **b** $y = -\frac{1}{2}x + 1$ **c** $y = 1 - x^2$ **d** $y = 5$

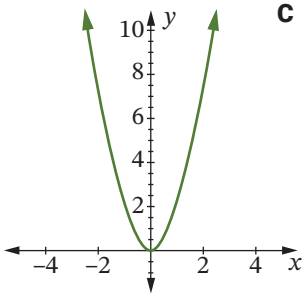
e $y = 3x^2 - 1$ **f** $y = 3^x$ **g** $y = -\frac{4}{x}$ **h** $y = 3^{-x}$

i $y = 2x^2$ **j** $y = 9x^2 - 4$ **k** $y = 3x - 2$ **l** $y = \frac{6}{x}$

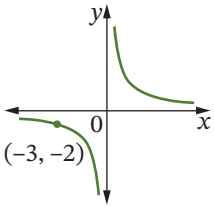
A



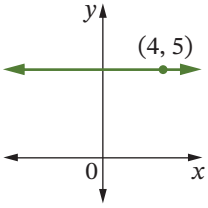
B



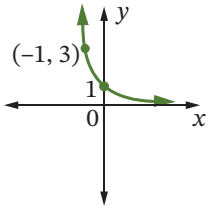
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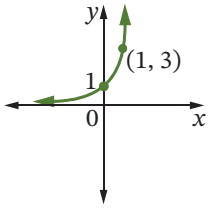
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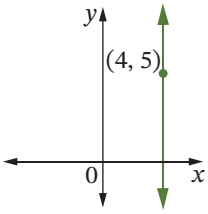
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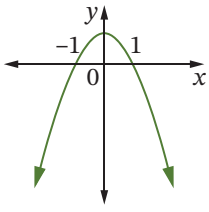
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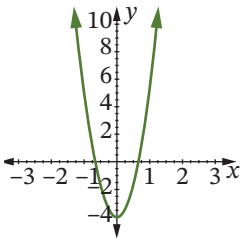
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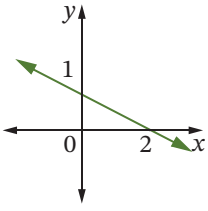
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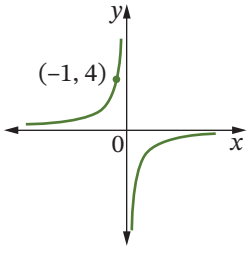
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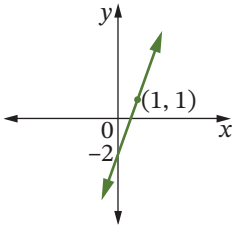
J



K



L



3 Sketch each equation, showing a point on the curve.

a $y = x^2 - 3$ **b** $y = 5^x$ **c** $y = -x^2 + 4$ **d** $y = 4 - x$

e $y = \frac{1}{2}x^2$ **f** $y = -2x + 4$ **g** $y = -\frac{5}{x}$ **h** $y = 10^{-x}$

4 Find the y-intercept of the graph of each equation.

a $y = 3^x$ **b** $y = 2x^2 + 3$ **c** $y = -7x^2 - 6$ **d** $y = 5^{-x}$

Foundation Standard Complex

Technology

Points of intersection

In this activity we will use graphing technology to find the point of intersection of a line and a curve.

1 Graph these equations of a line and parabola on the same screen.

$y = 3x - 4$ and $y = 2x^2 + x - 16$

2 How many points of intersection are there between the line and the parabola?

3 Click on each point of intersection to display its coordinates.

4 Graph each pair of equations to find the points of intersection of the line:

a $y = -x + 8$ and parabola $y = x^2 - 8x + 8$

b $x + y = 4$ and the exponential curve $y = 2^x$

c $y = 6x - 10$ and the parabola $y = 9x - x^2$

d $y = x - 3$ and hyperbola $y = -\frac{2}{x}$

e $y = x + 6$ and parabola $y = 5x^2$

f $15x + 8y = 17$ and exponential curve $y = 4^{-x}$

g $x - 2y = -6$ and hyperbola $y = \frac{8}{x}$

h $y = 2x - 1$ and hyperbola $y = \frac{15}{x}$

i $2x + y = -1$ and parabola $y = x^2 - 2x - 5$

j $x + 6y = 19$ and exponential curve $y = 3^x$

Technology

Identifying graphs

1 Use graphing technology to graph each equation and classify it as either a straight line (L), a parabola (P), a hyperbola (H) or an exponential curve (E).

a $y = 2x$ **b** $y = \frac{7}{x}$ **c** $y = x^2 + 1$

d $y = 2^x$ **e** $y = 2x^2 + 3$ **f** $y = 4 - 2x$

g $y = -\frac{1}{x}$ **h** $y = 2 - x$ **i** $y = 4 - x^2$

j $y = 5^{-x}$ **k** $y = -3x^2$ **l** $y = 3^x$

2 Without using graphing technology, classify each equation.

a $y = 3x - 2$ **b** $y = x^2 + 3$ **c** $y = 2^x + 1$

d $y = -\frac{4}{x}$ **e** $y = 4x^2 - 1$ **f** $y = 3^x - 2$

g $y = 4^x - 1$ **h** $y = 3x^2 - 4$ **i** $y = 10 - 2x^2$

j $y = -2x^2$ **k** $y = \frac{5}{x}$ **l** $y = -4 - 2x^2$

3 Check your answers to question **2** by drawing each equation using graphing technology.

4 State briefly in words how you distinguish between each type of equation in question **2**.

5 Use graphing technology to find the x-intercepts and y-intercepts (if they exist) of the graphs in questions **1** and **3**. Provide approximate answers where necessary.

Power plus ANSWERS ON P. 608

1 On the same set of axes, draw the graph of each equation.

a $y = x^2$ **b** $y = (x + 1)^2$ **c** $y = (x - 2)^2$

2 On the same set of axes, draw the graph of each equation.

a $y = -x^2$ **b** $y = -(x - 3)^2$ **c** $y = -(x + 2)^2$

3 For the graph of the parabola $y = (x + a)^2$, describe the effect on the graph of different values of a.

4 Draw the graph of each equation.

a $xy = 1$ **b** $y(x + 2) = 1$ **c** $y(x - 3) = 1$

CHAPTER 7 REVIEW

Quiz
Language of maths 7

Puzzle
Graphing curves crossword

Language of maths

asymptote	axis	coefficient	concave down
concave up	constant	conversion graph	curve
decay	direct variation	exponential	growth
hyperbola	inverse variation	parabola	proportional
quadratic	table of values	turning point	variable
vertex	vertical intercept	x-intercept	y-intercept

1 What is the coefficient of x^2 in the quadratic equation $y = 3x^2 + 10$?

2 What is the graph of $y = \frac{k}{x}$ called?

3 True or false: The exponential curve $y = 2^x$ passes through the point (0, 0).

4 In the variation equation $y = \frac{k}{x}$, which is the **constant of variation**?

5 Write down the equation of a parabola that is concave down and has a y-intercept of 3.

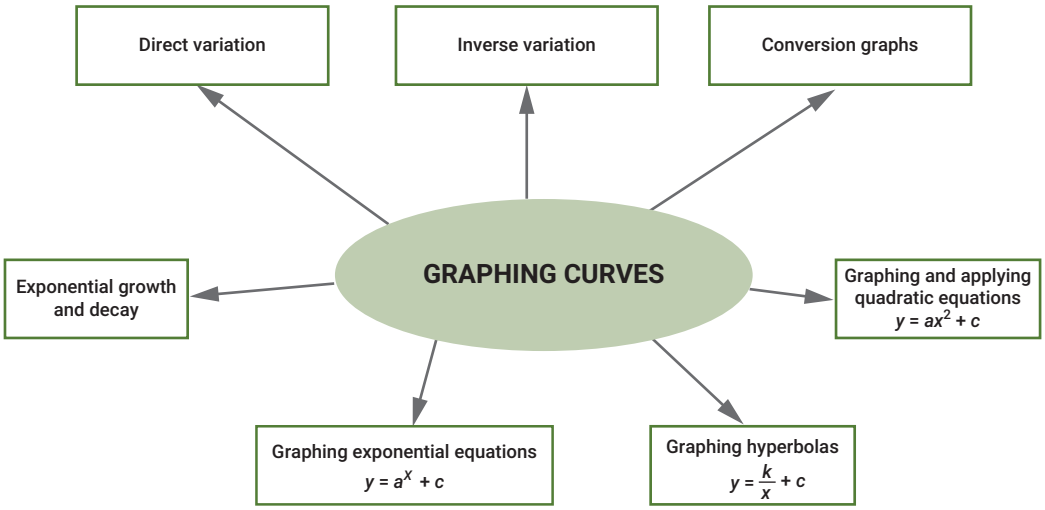
6 What is the asymptote of the exponential curve $y = a^x$?

Worksheet
Mind map: Graphing curves

Topic summary

- Which parts of this chapter were new to you?
- What is the difference between direct and inverse variation?
- Do you know the equations of a parabola, hyperbola and exponential curve, and how to graph them?
- Explain how the graph of $y = 2x^2 + 3$ is different from the graph of $y = -2x^2 + 3$. How are they similar?

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 7 ANSWERS ON P. 609

1 H is directly proportional to t. If when $t = 12$, $H = 138$, find H when $t = 27$.

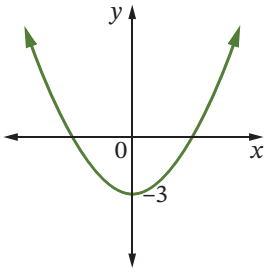
2 Air temperature, T (in degrees Celsius), varies inversely as its height, h (in metres), above sea level. At 400 m above sea level, the temperature is 15°C.
What is the temperature at 600 m above sea level?

3 The graph in Example **6** on page 293 converts Australian dollars to UK pounds sterling. Use the graph to convert:

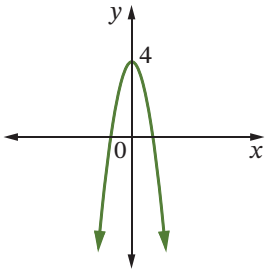
a \$A70 to £ **b** £56 to Australian dollars

4 Match each graph to its correct equation.

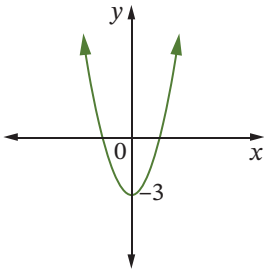
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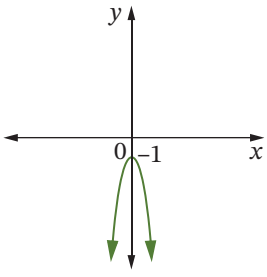
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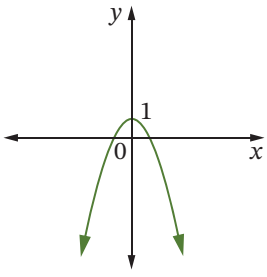
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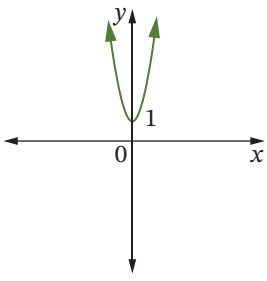
d



e



f



A $y = x^2 - 3$ **B** $y = 3x^2 + 1$ **C** $y = \frac{1}{2}x^2 - 3$

D $y = -x^2 + 1$ **E** $y = -4x^2 - 1$ **F** $y = 4 - 3x^2$

7.05

5 A stone is dropped from the top of a building and its height (h) in metres after t seconds is given by the quadratic equation $h = 70 - 4.9t^2$.

a Copy and complete this table of values and graph the equation.

Use a scale of 1 cm = 1 second on the horizontal axis and 1 cm = 10 m on the vertical axis.

Time (t seconds)	0	1	2	3	4
Height (h metres)					

b What is the vertical intercept and what does it represent?

c What is the height of the stone after 1.5 seconds?

d How far does the stone fall in the 3rd second?

e Use the graph to determine the time it takes for the stone hit the ground.

PATH

7.06

6 Graph each hyperbola, showing the coordinates of one point on the curve.

a $y = -\frac{3}{x}$ **b** $y = \frac{8}{x}$ **c** $y = \frac{2}{x+1}$ **d** $y = -\frac{4}{x}$

7.07

7 Graph each exponential equation.

a $y = 4^x$ **b** $y = 4^{-x}$ **c** $y = -4^x$ **d** $y = -4^{-x}$

7.08

8 Simone is given some money on her 5th birthday, and this amount is doubled each birthday until she turns 21. The amount of money ($\$A$) she receives each birthday is modelled by the exponential equation $A = 5(2^n)$, where n is the number of years after her 5th birthday.

a Copy and complete this table of values to graph the equation.

n (years)	0	1	2	3	4	6	8	10
A (cents)								

b How much money did Simone receive on her 5th birthday? How is this shown on your graph?

c How much did Simone receive on her 9th birthday (in dollars)?

d At what age will Simone receive more than $1500?

e How much will Simone receive on her 21st birthday?

7.08

9 The salvage value of a car ($\$V$) after n years can be calculated using the equation $V = 12\,000\,(1.24)^{-n}$.

a Copy and complete this table of values, correct to the nearest $10.

n	0	1	2	3	4	5
V						

b Use a scale of 1 cm = 1 year on the horizontal axis and 1 cm = $1000 on the vertical axis to graph the equation.

c What is the vertical intercept and what does it represent?

d How much does the value of the car drop in the:

i 1st year? **ii** the 4th year?

e Use guess-and-check to find when the value of the car falls below $1000.

10 Match each equation with its correct graph.

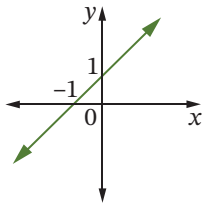
a $y = \frac{1}{4}x^2$ **b** $y = 3^x$ **c** $y = -2x^2 - 1$

d $x = -5$ **e** $y = -3x^2$ **f** $y = 2x^2 - 1$

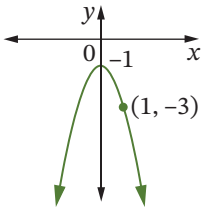
g $y = 3^{-x}$ **h** $y = -\frac{8}{x}$ **i** $y = x^2$

j $y = x + 1$ **k** $y = -5$ **l** $y = -2 - 2x$

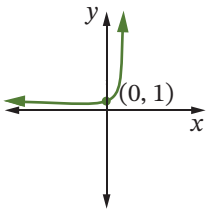
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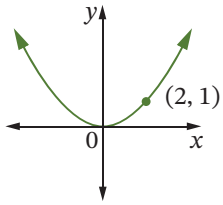
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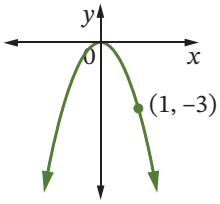
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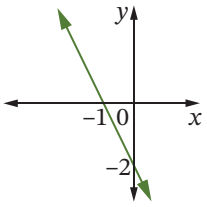
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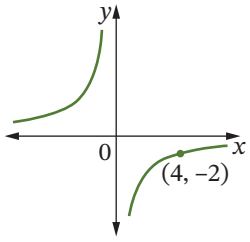
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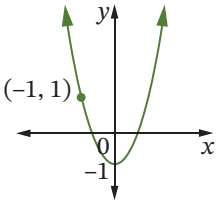
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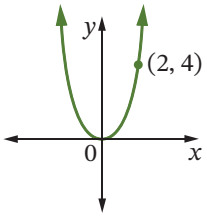
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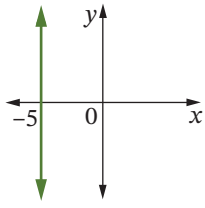
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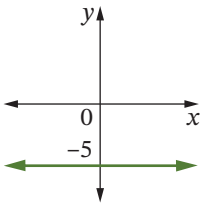
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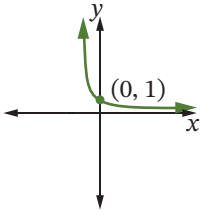
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K



L



Foundation | Standard | Complex

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